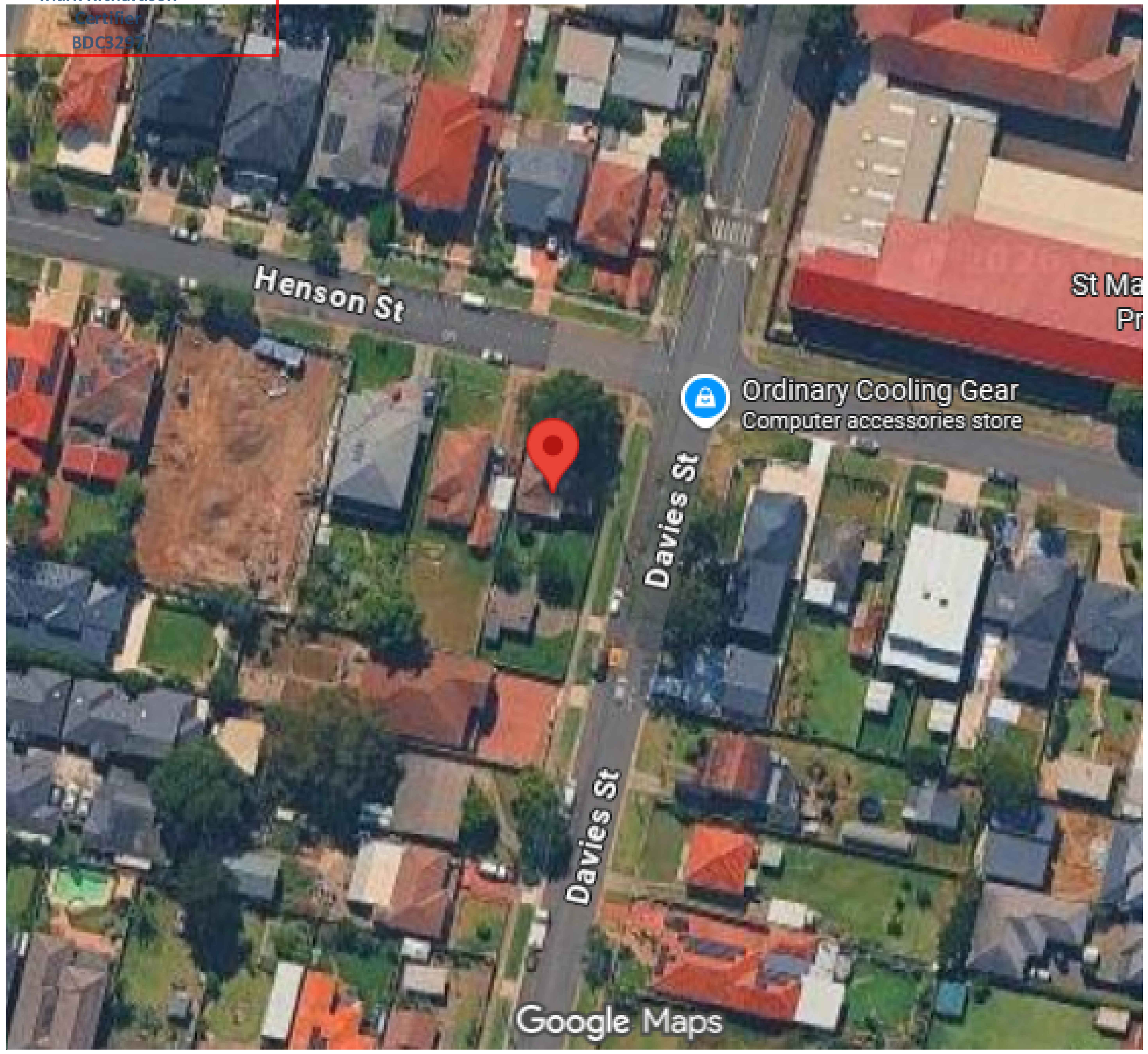
PROPOSED DUAL OCCUPANCY AT
16 HENSON STREET, MERRYLANDS, NSW 2160

MLR Certified Pty Ltd
Reference: 26-023/01
Date: 13/03/2026

Mark Richardson
Certifier
BDC3297



SITE LOCATION MAP

DRAWING LIST

SHEET No.	SHEET NAME
S1	COVER PAGE AND DRAWING LIST
S2	GENERAL NOTES
S3	CUT AND FILL PLAN
S4	POOL PLAN
S5	POOL PLAN SECTIONS & DETAILS
S6	GROUND FLOOR SLAB PLAN
S7	GROUND FLOOR SECTIONS & DETAILS
S8	GROUND FLOOR SECTIONS & DETAILS
S9	SEWER SECTION
S10	FIRST FLOOR AND LOWER ROOF FRAMING PLAN
S11	WALL BRACING PLANS
S12	ROOF FRAMING PLAN
S13	FRAMING DETAILS AND SECTIONS
S14	FRAMING DETAILS AND SECTIONS
S15	FRAMING DETAILS AND SECTIONS

NORTH POINT

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REVISIONS

Rev	Date	Description	Verified
REV A	13/02/2026	ISSUED FOR COORDINATION	J.M.

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GEMSTRUX
CONSULTING ENGINEERS

CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
COVER PAGE AND DRAWING LIST

APPROVED: J.M.
B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

PROJECT NO: G26043

DRAWING NO: S1

SCALE @ A1
NTS U.N.O

GENERAL NOTES

1. THESE ENGINEERING DRAWINGS ARE TO BE READ IN CONJUNCTION WITH PROJECT SPECIFICATIONS AND OTHER CONSULTANTS DRAWINGS ON THE PROJECT.
2. THESE ENGINEERING DRAWINGS HAVE BEEN PREPARED FROM INFORMATION AVAILABLE AT THE TIME OF ISSUE. AS THIS INFORMATION MAY BE THE SUBJECT OF CHANGE PRIOR TO OR DURING CONSTRUCTION THE CONTRACTOR IS TO ADVISE THE ENGINEER WHERE DISCREPANCIES OCCUR.
3. THESE DRAWINGS SHALL NOT BE USED FOR FINAL SETOUT OF THE PROJECT UNLESS SPECIFICALLY STATED.
4. INSPECTIONS ARE REQUIRED TO BE PERFORMED BY A DULY APPOINTED INSPECTOR FROM 'GEMSTRUX CONSULTING ENGINEERS'. THESE INSPECTIONS ARE REQUIRED TO BE PERFORMED IN ACCORDANCE WITH THE SCOPE OF INSPECTIONS PREPARED BY 'GEMSTRUX CONSULTING ENGINEERS'. THE INSPECTOR IS TO BE GIVEN A MINIMUM OF 48 HOURS NOTICE.
5. PRIOR TO THE COMMENCEMENT OF WORKS THE CONTRACTOR IS TO IDENTIFY ALL EXISTING SERVICES. ANY DAMAGE TO EXISTING SERVICES IS TO BE RECTIFIED AT THE CONTRACTORS EXPENSE. SERVICES SHOWN ON 'GEMSTRUX CONSULTING ENGINEERS' DRAWINGS ARE INDICATIVE ONLY.
6. DURING CONSTRUCTION THE CONTRACTOR SHALL BE RESPONSIBLE FOR MAINTAINING THE STABILITY OF THE WORKS AND ENSURE NO PART IS OVERSTRESSED.
7. WORKMANSHIP AND MATERIALS SHALL BE IN ACCORDANCE WITH THE CURRENT AUSTRALIAN STANDARDS AND BCA STATUTORY REQUIREMENTS, EXCEPT WHERE VARIED BY THE CONTRACT DOCUMENTS.
8. THE CONTRACTOR SHALL BE RESPONSIBLE FOR ENSURING THAT SUFFICIENT TOLERANCES ARE PROVIDED AND INTEGRATED THROUGHOUT ALL THE ELEMENTS OF THE WORKS.
9. ALL NON-LOAD BEARING ELEMENTS SHALL BE KEPT CLEAR OF THE STRUCTURE SOFFIT BY AN ALLOWANCE DETERMINED FROM SPAN/250 OR CANTILEVER/125 BUT NOT LESS THAN 20mm, UNLESS NOTED OTHERWISE ON THE DRAWINGS.
10. ALL DIMENSIONS ARE IN MILLIMETRES UNLESS STATED OTHERWISE.
11. ALL LEVELS ARE IN METRES (m) TO AUSTRALIAN HEIGHT DATUM AHD.
12. WIND AND EARTHQUAKE LOADS HAVE BEEN DETERMINED IN ACCORDANCE WITH AS1170.2 AND AS1170.4 RESPECTIVELY BASED ON THE FOLLOWING DESIGN CRITERIA :-

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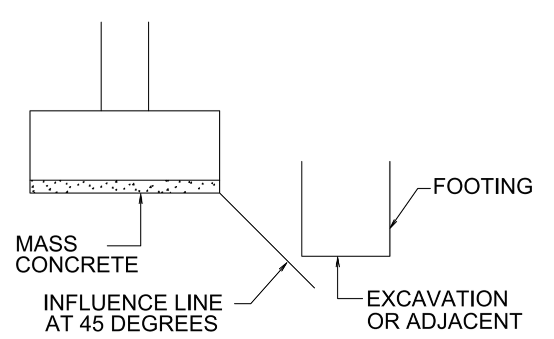
WIND LOADS:	
REGION	A2
TERRAIN CATEGORY	2.5
GUST WIND SPEED V_{zu}(m/s)	45
SHIELDING M_s	1.0
TOPGRAPHIC M_t	1.0
IMPORTANCE M_I	1.0

13. SUPERIMPOSED DEAD LOADS AND LIVE LOADS HAVE BEEN DETERMINED IN ACCORDANCE WITH AS1170.1 AND ARE SHOWN ON THE GENERAL ARRANGEMENT DRAWINGS.
14. ALL ABBREVIATIONS ARE IN ACCORDANCE WITH AS 1100. ADDITIONAL ABBREVIATIONS USED ARE AS FOLLOWS:

NSOP	NOT SHOWN ON PLAN
NSOE	NOT SHOWN ON ELEVATION
UNO	UNLESS NOTED OTHERWISE
U/S	UNDERSIDE
NGL	NATURAL GROUND LINE
FFL	FINISHED FLOOR LEVEL

FOUNDATION NOTES

1. REFER TO THE GEOTECHNICAL ENGINEERING REPORT SPECIFIED IN THE GENERAL NOTES FOR SITE SPECIFIC GEOTECHNICAL INFORMATION.
2. FOOTINGS TO BE FOUNDED ON MATERIAL HAVING AN ALLOWABLE BEARING CAPACITY OF 200 kPa ON SHALEY CLAY. WHERE DIFFICULTY IN REACHING THE REQUIRED CAPACITY IS EXPERIENCED, 'GEMSTRUX CONSULTING ENGINEERS' IS TO BE CONTACTED TO RE-ASSESS THE FOOTING DESIGN.
3. THE CONTRACTOR IS TO ENGAGE AND PAY A GEOTECHNICAL ENGINEER TO VERIFY THE BEARING CAPACITY OF THE FOUNDATIONS PRIOR TO PLACEMENT OF THE BLINDING LAYER.
4. ALL LOOSE MATERIAL AND WATER TO BE CLEANED OUT OF THE FOUNDATION. FORMWORK TO BE USED WHERE THE SIDES OF THE FOUNDATION ARE NOT STABLE.
5. A 50mm MINIMUM BLINDING LAYER SHOULD BE APPLIED TO THE BASE OF ALL FOUNDATIONS IMMEDIATELY AFTER VERIFICATION OF THE BEARING CAPACITY BY THE GEOTECHNICAL ENGINEER. WHERE THE FOUNDING MATERIAL IS DEEPER THAN REQUIRED FOR THE FOOTING THE EXCAVATION IS TO BE BACKFILLED WITH A WEAK MIX CONCRETE (N10) TO THE UNDERSIDE OF THE FOOTING.
6. WHERE AN EXCAVATION IS REQUIRED OR EXISTS BELOW THE BASE OF A FOOTING THE SIDE OF THE EXCAVATION SHALL BE LOCATED AWAY FROM EDGE OF FOOTING BY THE AME DISTANCE THAT THE EXCAVATION IS BELOW FOOTING BASE. WHERE THIS CANNOT BE ACHIEVED, 'GEMSTRUX CONSULTING ENGINEERS' SHALL BE CONTACTED FOR FURTHER DIRECTION. MASS CONCRETE IS TO EXTEND TO THE INFLUENCE LINE AS REQUIRED.
7. ALL WALLS AND COLUMNS SHALL BE CONCENTRIC WITH THE SUPPORTING FOOTINGS UNLESS NOTED OTHERWISE ON THE DRAWINGS.



CONCRETE NOTES

1. CONCRETE WORK SHALL BE IN ACCORDANCE WITH AS3600 AND WITH THE PROJECT SPECIFICATIONS.
2. CONSTRUCTION JOINTS SHALL BE PROPERLY FORMED AND USED ONLY WHERE SHOWN ON CONSULTING ENGINEERS'.
3. ALL THICKNESSES SHOWN ARE MINIMUM STRUCTURAL REQUIREMENTS, NO REDUCTION IN THICKNESS DUE TO FALLS OR TOPPING IS PERMITTED. REFER ARCHITECT DRAWINGS FOR ALL SLAB FALLS AND CONFIRMATION OF SLAB STEPS.
4. UNLESS A GROOVE LINE ALLOWANCE HAS BEEN NOTED ON THE DRAWINGS, NO GROOVE LINES ARE PERMITTED, EXCEPT AT SLAB LINES. ALL GROOVE LINES ARE TO BE SUBMITTED TO 'GEMSTRUX CONSULTING ENGINEERS' FOR APPROVAL.
5. THE FACE OF ALL CONCRETE AGAINST WHICH NEW CONCRETE IS TO BE CAST IS TO BE THOROUGHLY MECHANICALLY SCABBLED, FULLY EXPOSING THE AGGREGATE MATRIX.
6. NO PENETRATIONS GREATER THAN 150mm DIAMETER, OR EMBEDMENT OF PIPES GREATER THAN 40mm DIAMETER OTHER THAN THOSE SHOWN ON THE STRUCTURAL DRAWINGS SHALL BE MADE IN CONCRETE SLABS. FOR ALL OTHER CONCRETE MEMBERS NO PENETRATIONS, CHASES OR EMBEDMENTS SHALL BE MADE WITHOUT PRIOR APPROVAL BY 'GEMSTRUX CONSULTING ENGINEERS'.
7. CONDUITS GREATER THAN 25mm DIAMETER CAST INTO CONCRETE MEMBERS SHALL BE SPACED AT A MAXIMUM DISTANCE POSSIBLE AND UNDER NO CIRCUMSTANCES CLOSER THAN A CLEAR SPACING OF TWICE THE LARGER CONDUIT DIAMETER FROM PARALLEL REINFORCEMENT OR ANY OTHER CONDUIT.
8. THE CHARACTERISTIC COMPRESSIVE STRENGTH (f'c) AT 28 DAYS OF IN PLACE CONCRETE SHALL BE AS NOTED ON THE DRAWINGS.
9. MAXIMUM AGGREGATE SIZE.....20mm
10. SLUMP......................................80mm
11. ALL CONCRETE SHALL BE VIBRATED.
12. ALL CONCRETE SHALL BE CURED IN ACCORDANCE WITH THE SPECIFICATION.
13. ALL CONCRETE SHALL BE SAMPLED AND TESTED IN ACCORDANCE WITH AS1012 AND THE PROJECT SPECIFICATION.
14. ALL FORMWORK SHALL COMPLY WITH AS3610
15. EACH FLOOR SHALL BE FULLY PROPPED TO THE FLOOR BELOW IN ACCORDANCE WITH AS3610 (FORMWORK CODE).
16. THE FLOOR BELOW SHALL BE BACKPROPPED THROUGH A MINIMUM OF TWO STOREYS BELOW. THIS RESULTS IN A MINIMUM OF THREE STOREYS PROPPED AT ALL TIMES.
17. PROPS MAY BE REMOVED AFTER 28 DAYS OF CURING OR AFTER 14 DAYS IF THE CONCRETE HAS REACHED ITS CHARACTERISTIC STRENGTH (AS PROVED BY CYLINDER TEST RESULTS).

REINFORCEMENT

1. REINFORCEMENT IS TO BE MANUFACTURED IN ACCORDANCE WITH AS1302 AND SHALL BE FIXED AS SHOWN ON DRAWINGS.
2. MATERIAL IS INDICATED BY THE FOLLOWING SYMBOLS:-

   Y DEFORMED BAR GRADE 400
   N DEFORMED BAR GRADE 500 (NORMAL DUCTILITY)
   R PLAIN ROUND BAR GRADE 250
   W PLAIN WIRE GRADE 450
   F FABRICS GRADE 450

3. THE BAR SIZE IS INDICATED BY A NUMBER AFTER THE SYMBOL, WHICH INDICATES THE BAR DIAMETER IN MILLIMETERS.
4. REINFORCEMENT SPACING NOMINATED ON DRAWINGS IS TO ASSIST SCHEDULER AND STEELFIXER TO ASSESS TOTAL NUMBER OF BARS REQUIRED. WHERE BARS PLACED IN ACCORDANCE WITH SPACING NOMINATED FOUL WITH OTHER STRUCTURAL REQUIREMENTS, PREFERANCE IS TO BE GIVEN TO RELOCATING BARS BY LOCALLY ADJUSTING SPACING TO ENABLE ASSEMBLY OF REINFORCEMENT TO BE COMPLETED. ENGINEER IS TO BE CONTACTED IN THE EVENT THAT REINFORCEMENT IS NEEDED TO BE CUT ON SITE PROIR TO CONTINUING.
5. LAP LENGTHS TO REINFORCEMENT BARS TO BE AS NOTED ON THE RELEVANT DRAWINGS.
6. WELDING OF REINFORCEMENT BARS IS NOT PERMITTED UNLESS APPROVED.
7. COVER SHALL BE AS NOTED ON THE RELEVANT DRAWINGS.
8. CONCRETE COVERS NOTED ARE MEASURED FROM THE FORMWORK OR GROUND FACE TO THE OUTERMOST REINFORCEMENT COMPONENT. ie. IN COLUMNS AND BEAMS TO THE OUTSIDE OF TIES OR LIGATURES.
9. COVER TO BE MAINTAINED DURING POURING BY THE USE OF PLASTIC CHAIRS OR PLASTIC TIPPED METAL CHAIRS.
10. WHERE NO REINFORCEMENT IS SHOWN ON THE DRAWING AT RIGHT ANGLES TO THE MAIN REINFORCEMENT DISTRIBUTION REINFORCMENT IS TO BE PROVIDED.

MASONRY NOTES

1. ALL MASONRY SHALL COMPLY WITH AS3700 AND THE PROJECT SPECIFICATION.
2. CONCRETE MASONRY UNITS TO HAVE A MINIMUM CHARACTERISTIC UNCONFINED STRENGTH OF 20 MPa IN ACCORDANCE WITH AS2733.
3. MASONRY TO BE BEDDED IN FRESHLY PREPARED MORTAR.

(a) CONCRETE BLOCKS :-
MORTAR MIX TO BE UNIFORMLY MIXED IN A RATIO OF ONE PART CEMENT, ONE PART LIME AND SIX PARTS SAND CONFORMING TO AS2701. 'BRICKIES LOAM' SHALL NOT BE USED.

(b) CLAY BRICKS :-
MORTAR MIX TO BE UNIFORMLY MIXED IN THE RATIO OF ONE PART CEMENT, THREE PARTS SAND AND ONE FOURTH PART LIME CONFORMING TO AS2701. 'BRICKIES LOAM' SHALL NOT BE USED.

4. GROUT SHALL HAVE A COMPRESSIVE STRENGTH (f'c) OF 20 MPa AT 28 DAYS, A SLUMP OF 125mm IN A 150mm SLUMP CONE, A MAXIMUM AGGREGATE SIZE OF 10mm AND BE IN ACCORDANCE WITH AS3700.
5. BEDDING OF MASONRY SHALL BE FULL FACE WITH CROSS JOINTS COMPLETELY FILLED. JOINT THICKNESS SHALL NOT EXCEED 12mm.
6. PROVIDE WALL TIES AT 600mm MAXIMUM CENTRES VERTICALLY AND HORIZONTALLY. REFER TO MASONRY DETAILS FOR WALL TIE SETOUT AT OPENINGS.
7. THE CAVITY SHALL NOT EXCEED 100mm AND SHALL NOT BE SMALLER THAN 40mm UNLESS NOTED OTHERWISE. KEEP CAVITY CLEAN AND CLEAR OF OBSTRUCTIONS.
8. RAKING OF JOINTS IS NOT PERMITTED WITHOUT PRIOR APPROVAL FROM 'GEMSTRUX CONSULTING ENGINEERS.
9. ALL WALLS TO BE KEPT STABLE AT ALL STAGES OF CONSTRUCTION AND NOT BE OVER STRESSED AT ANY TIME.
10. PERMITTED IN MASONRY WALLS WITHOUT THE PRIOR APPROVAL OF 'GEMSTRUX CONSULTING ENGINEERS'.

TIMBER NOTES

T1 AS 1684 IS RELEVANT TO DOMESTIC CONSTRUCTION IN SHELTERED LOCATIONS.

T2 SOFTWOOD MINIMUM GRADE F7 U.N.O.
HARDWOOD MINIMUM GRADE F11 U.N.O.

T3 EXTERNAL TIMBER TO BE EITHER HARDWOOD
DURABILITY CLASS I OR II OR IMPREGNATED
GRADE F7. PRESSURE TREATED TO AS1684 AND RE-DRILLED PRIOR TO USE.
SUPPLEMENTARY TREATMENT
SHALL BE APPLIED TO ALL CUT SURFACES.
PROVIDE DOCUMENTATION.

T4 ALL BOLTS IN TIMBER CONSTRUCTION TO BE MIN.
M16 U.N.O. BOLT HOLES TO BE DRILLED EXACT SIZE.
WASHERS UNDER HEADS AND NUTS TO BE AT LEAST 2.5 TIMES BOLT DIAMETER.

T5 FINISHED TIMBER SIZES.
SEASONED SOFTWOOD +5,-0mm
UNSEASONED SOFTWOOD F7+3,-3mm
F7+2,-4mm
SEASONED HARDWOOD +2,-0mm
UNSEASONED HARDWOOD -3,-3mm
(SEE ALSO CLAUSE 1.6.2 IN AS 2082)

T6 ALL TIMBER JOINTS AND NOTCHES TO BE 100mm MINIMUM FROM LOOSE KNOTS.
SEVERE SLOPING GRAIN, GUM VEINS OR OTHER MINOR DEFECTS.
FOR JOISTS SPANNING GREATER THAN 3m AND LESS THAN 4.2m
PROVIDE ONE ROW OF BLOCKING MID-SPAN.
FOR JOISTS SPANNING GREATER THAN 4.2m AND UP TO 6.0m
PROVIDE TWO ROWS OF BLOCKING AT 1/3 POINTS.
FOR DEEP JOISTED FLOORS WHERE A CONTINUOUS TRIMMING JOIST
IS NOT PROVIDED AT END OF JOISTS.
BLOCKING IS REQUIRED AT 1800 MAXIMUM CENTERS. (REFER TO AS 1684)

T7 BLOCKING IS NOT REQUIRED FOR JOISTS SPANNING LESS THAN 3m.

STRUCTURAL STEELWORK NOTES

S1 ALL MATERIALS, WORKMANSHIP, FABRICATION AND ERECTION SHALL COMPLY WITH THE REQUIREMENTS OF AS4100, AS1538, AS1554 AND THE SPECIFICATION.

S2 UNLESS SHOWN OTHERWISE, ALL STEEL SHALL BE IN ACCORDANCE WITH AS1204 GRADE 300. ALL STEEL HOLLOW SECTIONS SHALL BE GRADE 350 U.N.O. AND SHALL BE IN ACCORDANCE WITH AS1163.ALL PRESSED METAL PURLINS AND GIRTS SHALL BE GRADE 450 STEEL IN ACCORDANCE WITH AS1538

S3 UNLESS SHOWN OTHERWISE ON THE DRAWINGS, ALL CONNECTIONS SHALL BE IN ACCORDANCE WITH THE FOLLOWING MINIMUM REQUIREMENTS:

ALL WELDS SHALL BE 6MM CONTINUOUS FILLET WELDS ALL ROUND.
ALL BOLTS SHALL BE M20 - 8.8/S, WITH A MINIMUM OF 2 BOLTS PER CONNECTION.
PURLIN BOLTS TO BE M12 - 4.6/S WITH A MINIMUM OF 2 BOLTS PER PURLIN END.
ALL GUSSET AND CLEAT PLATES SHALL BE 10mm THICK. (U.N.O.)
ALL CAP PLATES SHALL BE 10 mm THICK. (U.N.O.)
ALL BASE PLATES SHALL BE 10 mm THICK. (U.N.O.)

S4 BOLT DESIGNATION:

4.6/S REFERS TO COMMERCIAL BOLTS OF STRENGTH GRADE 4.6 TO AS1111 TIGHTENED TO A SNUG TIGHT CONDITION.

8.8/S REFERS TO HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 TIGHTENED TO A SNUG TIGHT CONDITION.

8.8/TB REFERS TO HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 FULLY TENSIONED TO AS4100 AS A BEARING JOINT.

8.8/TF REFERS TO HIGH STRENGTH STRUCTURAL BOLTS OF GRADE 8.8 TO AS1252 FULLY TENSIONED TO AS4100 AS A FRICTION JOINT.

S5 HIGH STRENGTH BOLTED JOINTS SHALL BE IN ACCORDANCE WITH AS1511. THE SPECIFIED BOLT TENSION SHALL BE OBTAINED BY USE OF THE "PART TURN" METHOD OF TIGHTENING.

S6 ALL WELDS SHALL BE SP (SPECIAL PURPOSE) IN ACCORDANCE WITH AS1554. ALL ELECTRODES SHALL BE CLASS E48. ALL BUTT WELDS SHALL BE FULL STRENGTH COMPLETE PENETRATION WELDS.

S7 SUBSTITUTIONS FOR STEEL SECTIONS SHOWN ON DRAWINGS SHALL NOT BE MADE WITHOUT THE APPROVAL OF THE ENGINEER.HAN 3m.

S8 ALL STEELWORK BELOW GROUND OR FINISHED SURFACE LEVEL IS TO BE HOT-DIP GALVANIZED.

DESIGN SPECIFICATIONS	
SITE CLASSIFICATION REPORT	
COMPANY	GEMSTRUX CONSULTING ENGINEERS
REPORT NUMBER (REF)	SC26004
DATE	15 JANUARY 2026
SITE DETAILS	
SITE CLASSIFICATION	P
SAFE WORKING CAPACITIES	
FOOTINGS	100 kPa
PIERS	200 kPa

NORTH POINT

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B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

PROJECT NO:
G26043

DRAWING NO:
S2

SCALE @ A1
NTS U.N.O

CUT & FILL PLAN

SCALE 1:100

NOTES:

1. CUT AND FILL PLAN SHOWN IS INDICATIVE ONLY AND PROVIDED FOR STRUCTURAL DESIGN PURPOSES. BUILDER/CONTRACTOR IS RESPONSIBLE FOR DETERMINING AND SETTING OUT BULK EXCAVATION LEVELS IN ACCORDANCE WITH THE ARCHITECTURAL DRAWINGS, SITE CONDITIONS, AND THEIR CONSTRUCTION METHODOLOGY.

LEGEND

(hatched)	DENOTES CUT
(dotted)	DENOTES FILL

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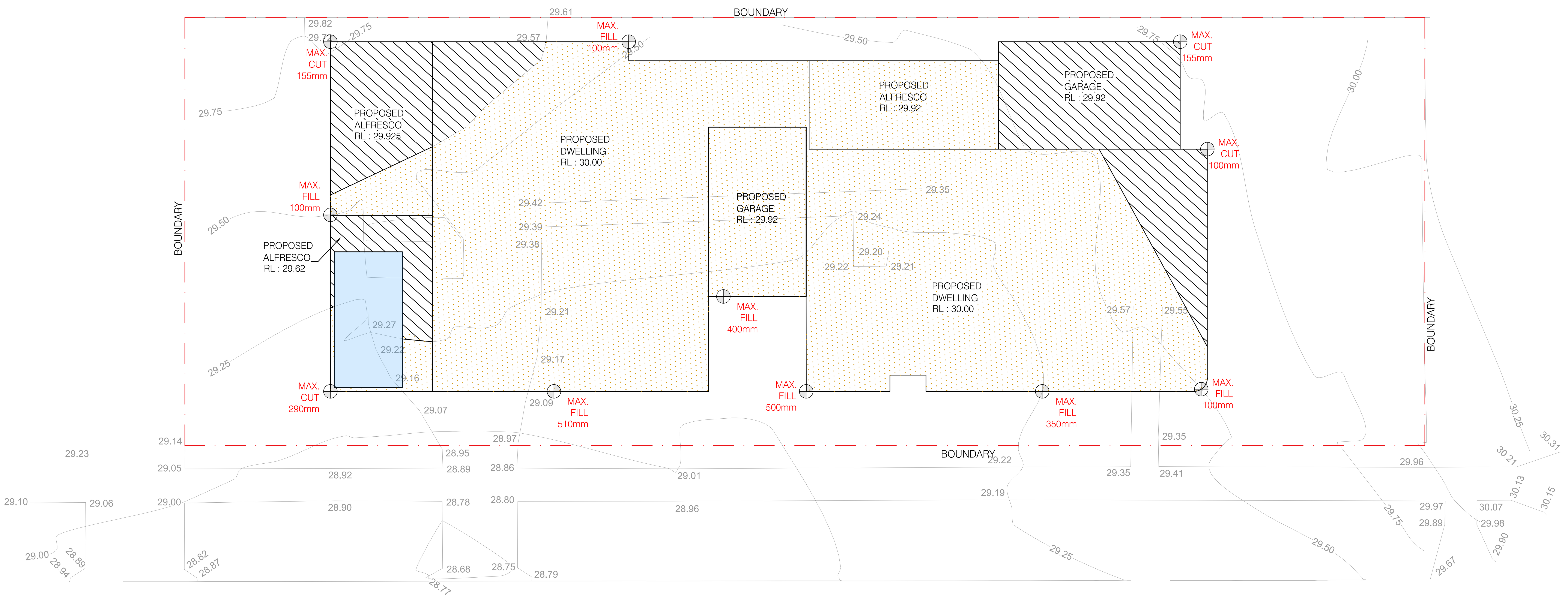

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CUT AND FILL PLAN

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MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

0 1 2 3 4 5 10
GRAPHIC SCALE: 1:100

PROJECT NO:
G26043

DRAWING NO:
S3

SCALE @ A1
1:100 U.N.O

POOL

SCALE 1:100

NOTE :

1. SITE IN ACCORDANCE WITH AS2870 CLASS P- AS PER SITE CLASSIFICATION REPORT PREPARED BY GEMSTRUX CONSULTING ENGINEERS (REPORT REFERENCE: SC26004), DATED 15 JANUARY 2026.
2. ALL EXPOSED SLABS TO BE WATERPROOFED WITH APPROVED SYSTEMS
3. DRAWING TO BE READ IN CONJUNCTION WITH ARCHITECTURALS
4. REFER TO ARCHITECTURAL DRAWINGS FOR ALL SETDOWNS, LEVELS, FALLS ETC.
5. I.J. - DENOTES ISOLATION JOINST - 10mm POLYETHYLENE (MUST BE LOCATED BETWEEN THE WALLS/COLUMNS AND THE NEW SLAB)

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CONSTRUCTION NOTES

1. ALL DIMENSIONS LOCATING POOL ARE TO BE TAKEN FROM ARCHITECTURAL DRAWINGS
2. WRITTEN DIMENSIONS TO BE TAKEN IN REFERENCE TO SCALE
3. ENGINEER TO BE ADVISED IF EXCAVATION IS IN FILL OR EXCESSIVE GROUND WATER IS ENCOUNTERED
4. SUPPORTING FOUNDATION MATERIAL TO BE SHALEY CLAY OF UNIFORM MOISTURE CONTENT WITH SAFE BEARING CAPACITY OF 200KPa
5. WHERE IT IS CONSIDERED THAT GROUND WATER CAN BUILD UP TO A LEVEL 500mm ABOVE THE FLOOR OF THE EXCAVATION ADEQUATE DRAINAGE SHALL BE PROVIDED UNDER THE POOL FLOOR
6. CONCRETE TO HAVE A MINIMUM DESIGN STRENGTH OF f'c = 32MPa AT 28 DAYS CONCRETE TO BE PNEUMATICALLY APPLIED
7. UPON COMPLETION OF CONCRETING THE HYDROSTATIC VALVE IS TO BE CHECKED TO ENSURE EFFECTIVE AND SUFFICIENT OPERATION
8. ALL REINFORCEMENT TO BE SUPPORTED BY BAR CHAIRS
9. REINFORCEMENT IS TO BE STRUCTURAL GRADE DEFORMED BAR GRADE TO 230 TO AUSTRALIAN STANDARD AS1302
10. WATER FACE REINFORCEMENT TO HAVE 65mm CONCRETE COVER REAR FACE REINFORCEMENT HAVE 50mm COVER FROM REAR REAR FACE IF FORMED AND 65mm COVER ISF SPRAYED AGAINST GROUND
11. ALL BARS SHALL BE SPLICED 40 BAR DIAMETERS MIN.
12. SPLICES IN BOND BEAM BARS SHALL BE STAGGERED
13. ALTERNATIVE REINFORCEMENT TO BE TEMCORE BARS IN ACCORDANCE WITH AS 1302 410 Y

LEGEND

	BRICKWORK ABOVE
180	SLAB THICKNESS
P1	450 Ø MASS CONCRETE PIER TO BE BEARING 1.5m INTO 200 kPa SHALEY CLAY
P2	450 Ø REINFORCED PIER TO BE BEARING 1.5m INTO 200 kPa SHALEY CLAY
IJ	ISOLATION JOINT - 10mm FILLER BOARD

MESH SCHEDULE

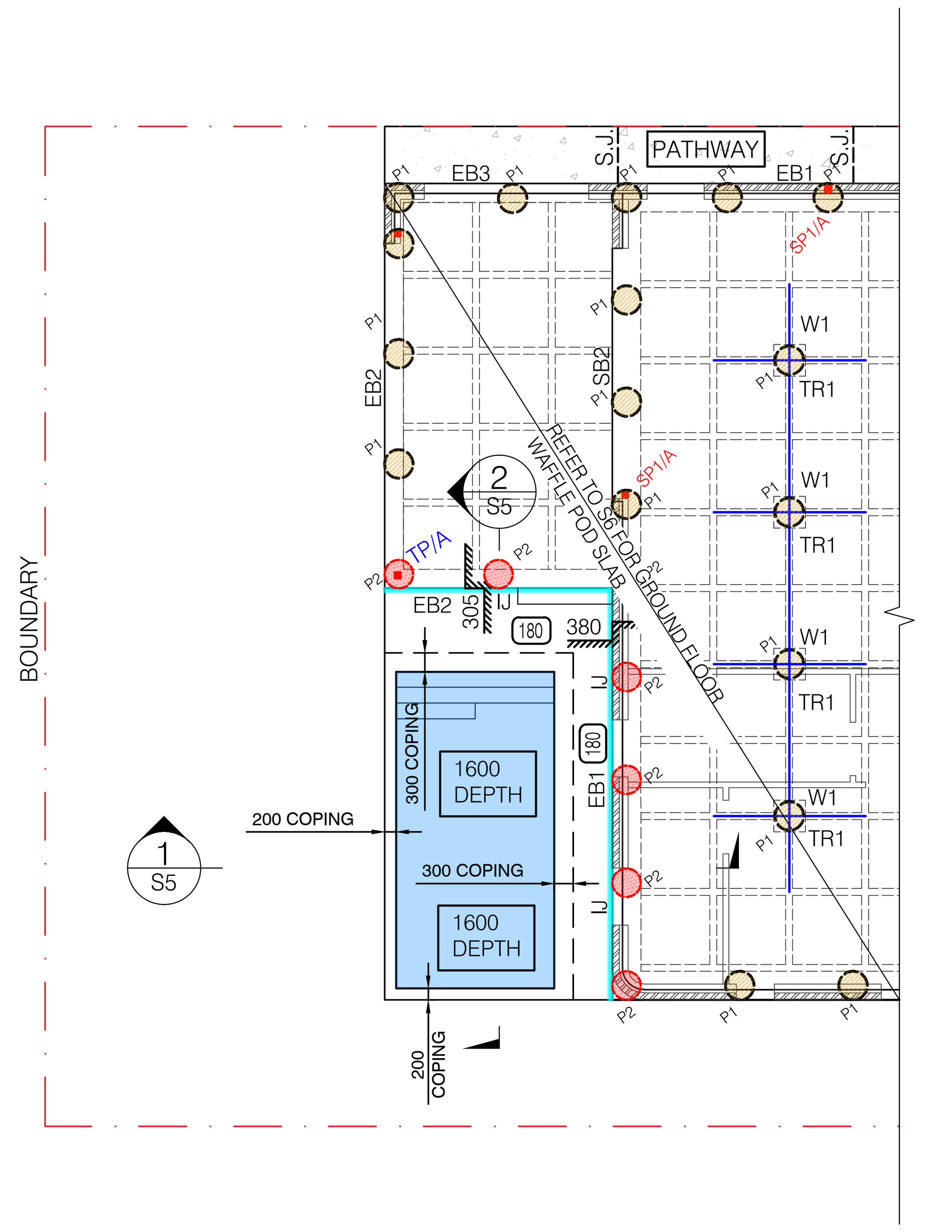
SLAB THICKNESS	BOTTOM REINFORCEMENT	TOP REINFORCEMENT
180	SL82 MESH	SL82 MESH

CONCRETE QUALITY

ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f 'c
SLAB ON GROUND	80mm	20mm	A	32MPa
SWIMMING POOL	80mm	20mm	A	32MPa
FOOTING	80mm	20mm	A	32MPa

REINFORCEMENT COVER SCHEDULE

MEMBER	COVER (mm)			EXPOSURE CLASSIF.
	TOP	BOTTOM	SIDES	
INTERNAL SLABS	30mm	30mm	30mm	A1
EXTERNAL SLABS	40mm	40mm	40mm	A2
FOOTING	40mm	40mm	40mm	A2



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DRAWING NO:
S4

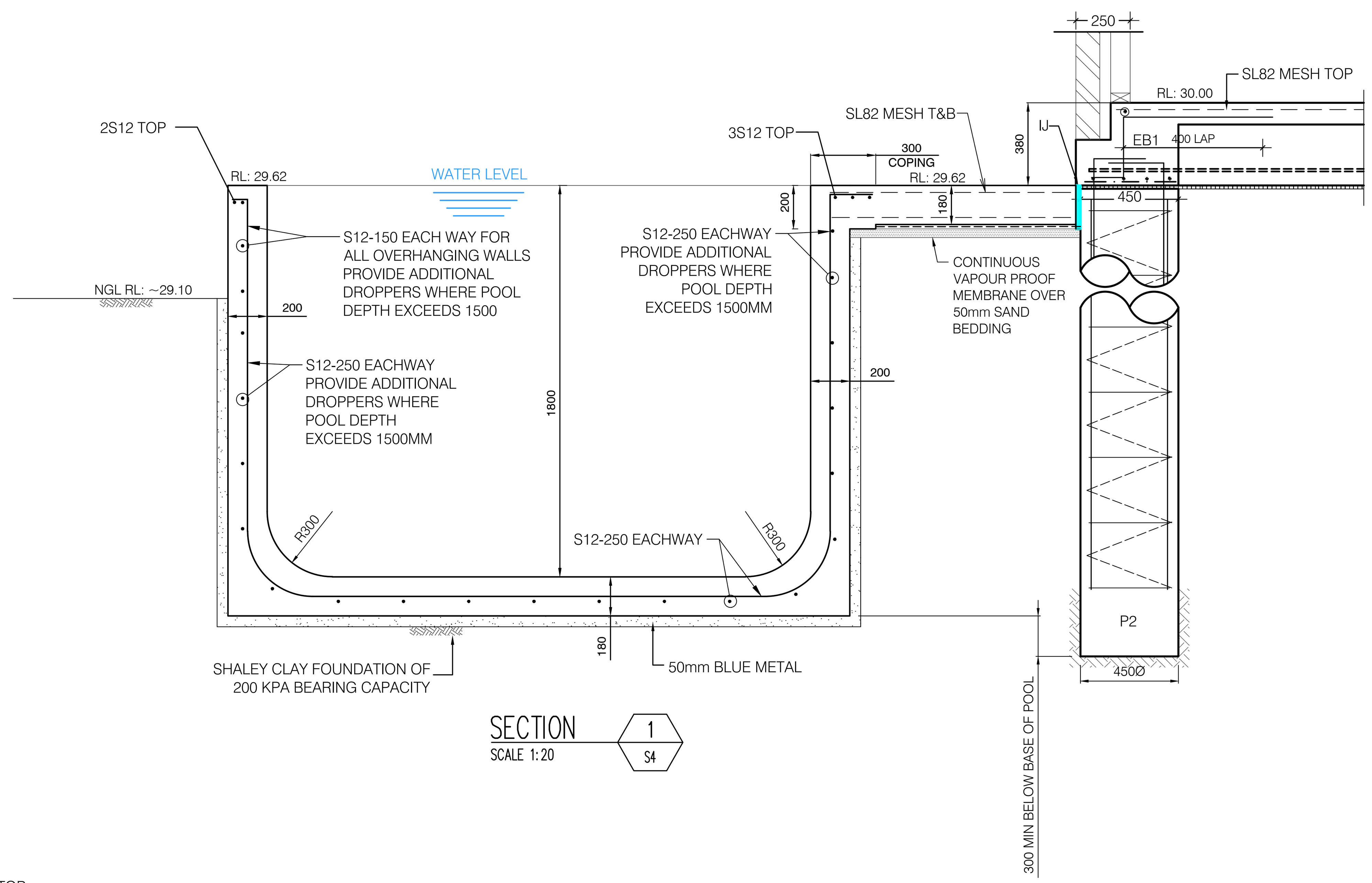
0 1 2 3 4 5 10

GRAPHIC SCALE: 1:100

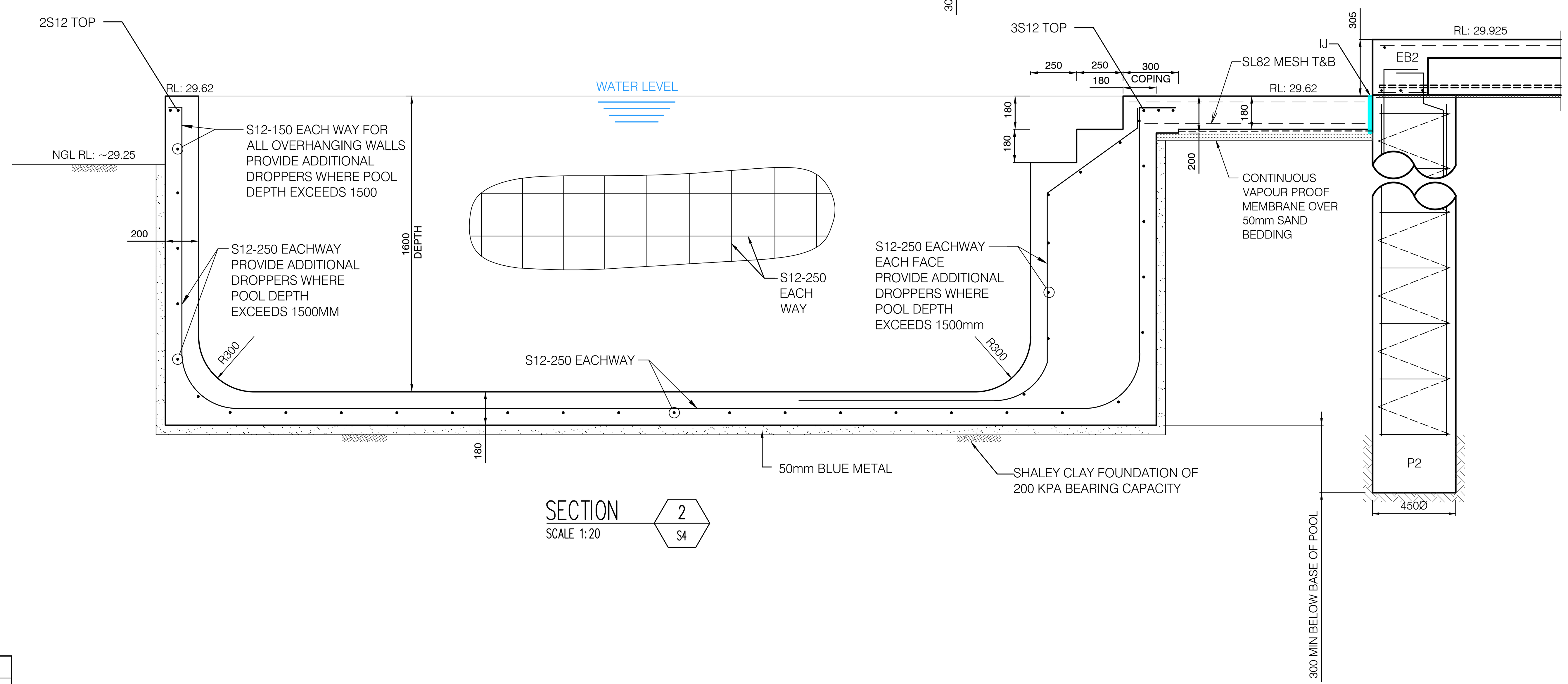
SCALE @ A1
1:100 U.N.O

MLR Certified Pty Ltd
Reference: 26-023/01
Date: 13/03/2026

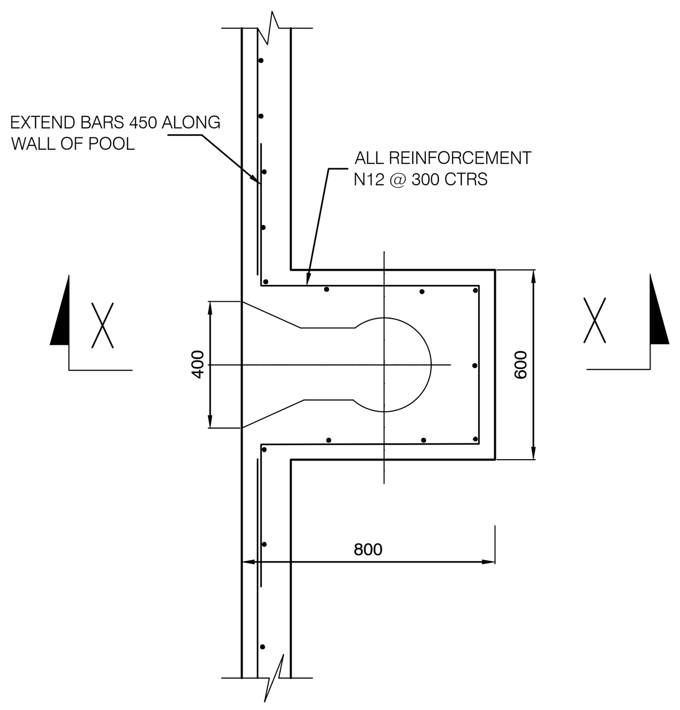
Mark Richardson
Certifier
BDC3297



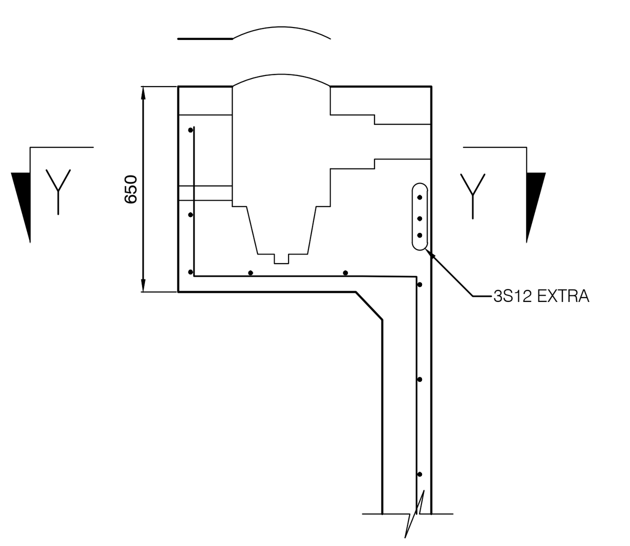
SECTION 1 / S4
SCALE 1:20



SECTION 2 / S4
SCALE 1:20

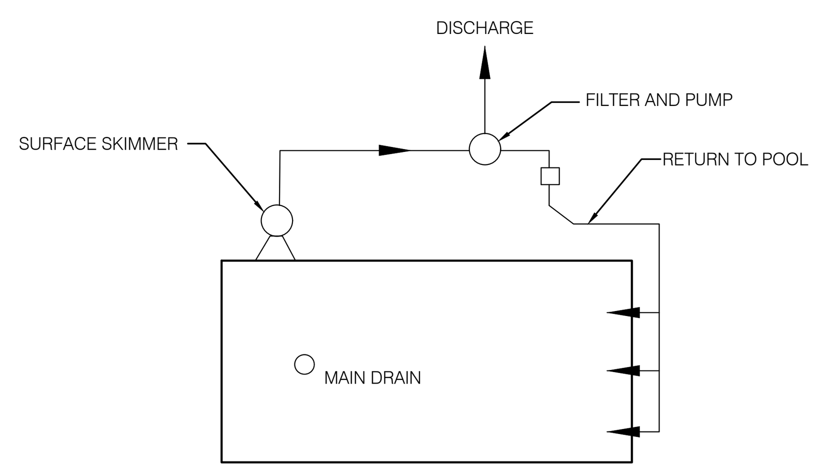


PLAN - SECTION Y-Y



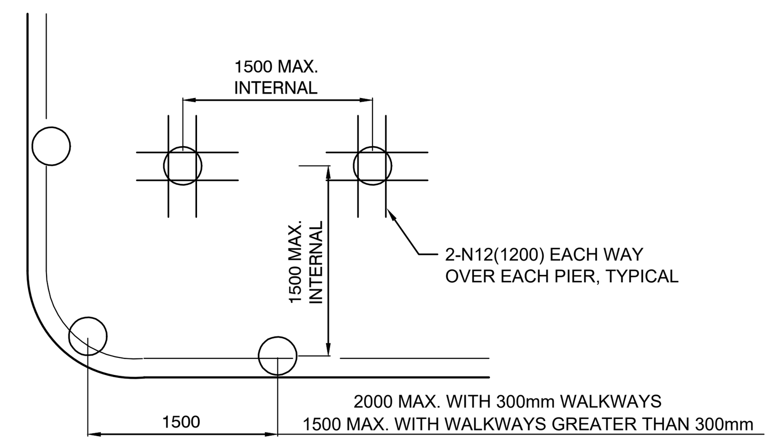
ELEVATION - SECTION X-X

SKIMMER BOX DETAIL



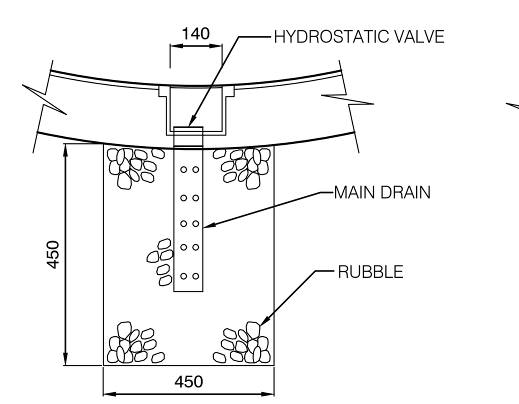
PLUMBING LAYOUT

DIAGRAMATIC ONLY



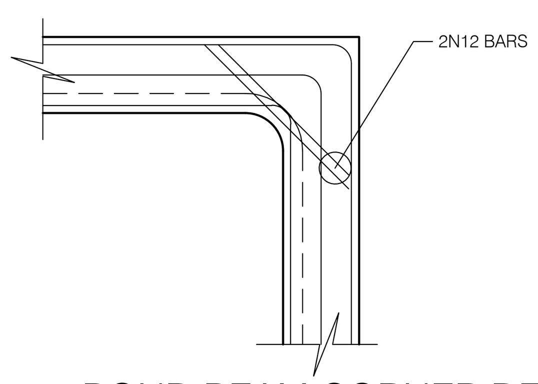
POOL PIER LAYOUT DETAIL

SCALE 1:50



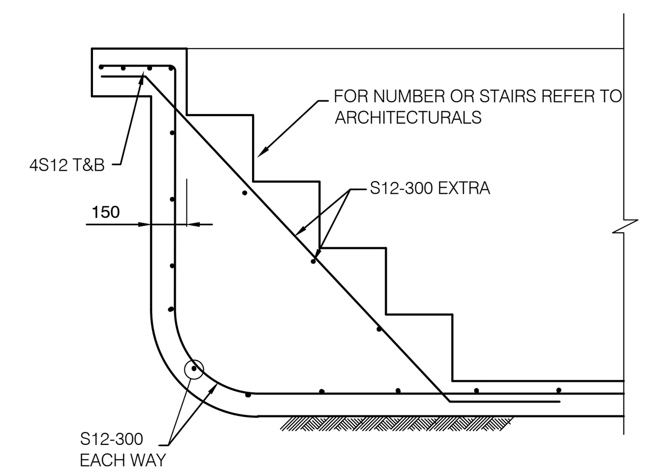
MAIN DRAIN DETAIL

SCALE 1:20



BOND BEAM CORNER DETAIL

SCALE 1:50



TYPICAL STAIR SECTION

NORTH POINT

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REVISIONS

Rev	Date	Description	Verified
REV A	13/02/2026	ISSUED FOR COORDINATION	J.M.

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GEMSTRUX
CONSULTING ENGINEERS

CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
POOL PLAN
SECTIONS & DETAILS

APPROVED: J.M.
B E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

0 0.1 0.2 0.3 0.4 0.5 1.0 1.5 2.0
GRAPHIC SCALE 1:20

PROJECT NO:
G26043

DRAWING NO:
S5

SCALE @ A1
1:20 U.N.O

GROUND FLOOR - WAFFLE POD PLAN - 100 mm SLAB U.N.O

SITE IN ACCORDANCE WITH AS2870 CLASS P - AS PER SITE CLASSIFICATION REPORT PREPARED BY GEMSTRUX CONSULTING ENGINEERS (REPORT REFERENCE: SC26004), DATED 15 JANUARY 2026.

NOTE :

1. f'c = 32 MPa INTERNAL AREAS
2. 100mm THICK SLAB SL82 TOP THROUGHOUT
3. 150mm THICK SLAB SL82 TOP & BTM THROUGHOUT
3. ALL INTERNAL PODS 300mm, GARAGE & PATIO PODS 225mm
4. ALL BRICKS WALLS TO BE ARTICULATED, EXPANSION JOINT SPACING TO BE 6000mm
5. ALL CORNERS TO HAVE L-BAR WITH 500mm LEGS TOP AND BOTTOM
6. PODS LESS THAN 400mm LONG CAN BE TAPED
7. PIERS TO BE POURED SEPARATE TO WAFFLE POD RAFT SLAB
8. CONCRETE PIERS TO BE f'c = 32 MPa
9. ALL EXPOSED SLABS TO BE WATERPROOFED WITH APPROVED SYSTEMS
10. SAFE BEARING CAPACITY HAS BEEN TAKEN TO BE A MIN. OF 200 kPa TO BE BEARING ON SHALEY CLAY

MLR Certified Pty Ltd
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Mark Richardson
Certifier
BDC3297

150mm THICK SLAB WITH SL82 MESH TOP & BOTTOM TO BE USED FOR ALL SLABS 700mm ABOVE N.G.L.

DRIVEWAY- 120mm SLAB WITH SL82 MESH TOP U.N.O

NOTE :1. MAX 6m SAWN CONTROL JOINT
2. MAX 15m KEYED CONSTRUCTION JOINT
3. MAX 30m EXPANSION JOINT

PATHWAY- 100mm SLAB WITH SL72 MESH TOP U.N.O

NOTE :1. MAX 4m SAWN CONTROL JOINT
2. MAX 10m KEYED CONSTRUCTION JOINT
3. MAX 20m EXPANSION JOINT

LEGEND

	BRICKWORK ABOVE
	STUD WALLS ABOVE
	STARTING POD
P1	450Ø MASS CONCRETE BORED PIERS - BEARING 1.5m INTO 200 kPa SHALEY CLAY
P2	450Ø REINFORCED CONCRETE BORED PIERS - BEARING 1.5m INTO 200 kPa SHALEY CLAY
P1 W1	500 x 500 MASS CONCRETE PAD CUT-OUT
TR	2N12 TRIMMERS TOP - 1200 LONG
TR1	DENOTES N12 TRIMMER BARS - 2400 LONG EACH WAY
S.J.	SAWN CUT JOINT
K.J.	KEY JOINT (REFER TO DETAIL)
IJ	ISOLATION JOINT - 10mm FILLER BOARD
75	STEP DOWN IN SLAB
TP/A	DENOTES 90x90 HARDWOOD POST ABOVE
SP/A	DENOTES STEEL POST ABOVE (89x89x6.0mm SHS)- HOT DIP GALV. (IF EXTERNAL OR EXPOSED)
	DENOTES BRIDGE FOOTINGS

REINFORCEMENT COVER SCHEDULE

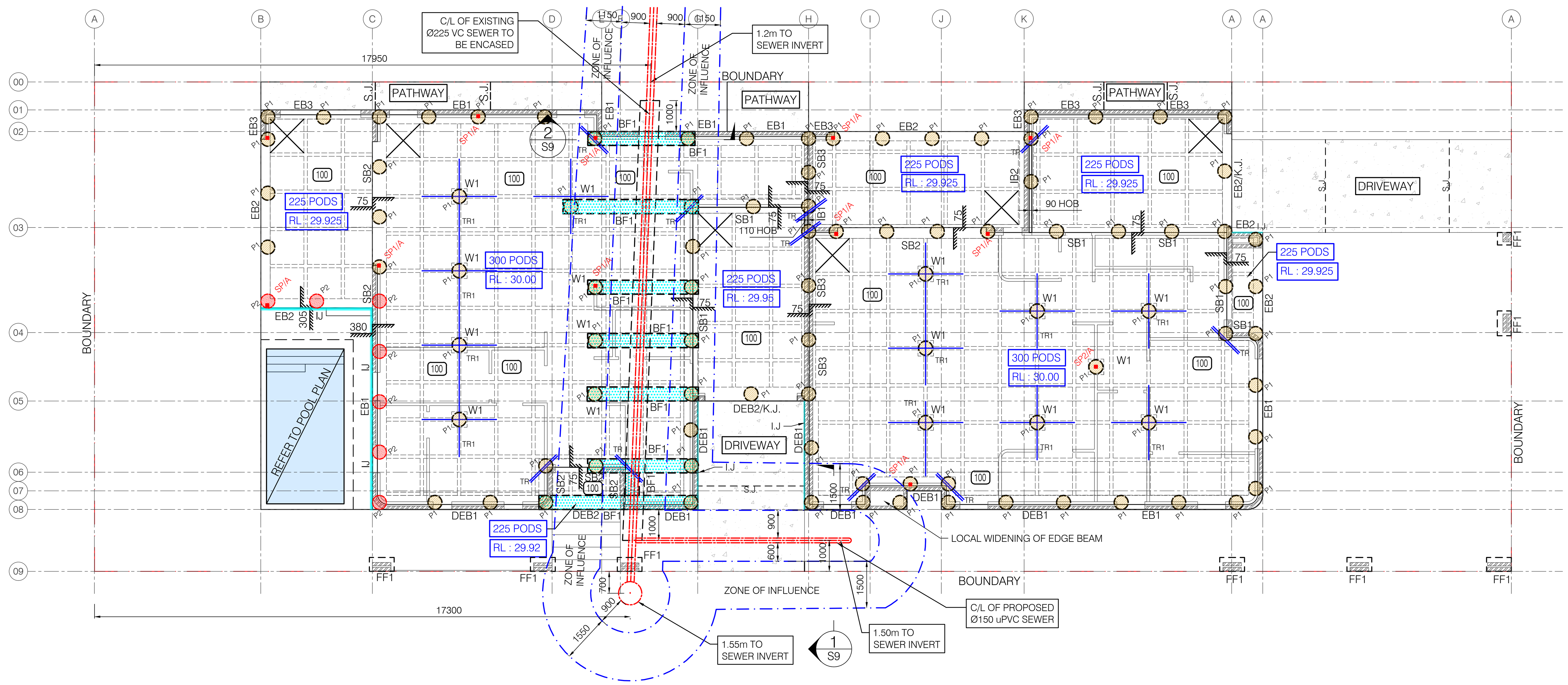
MEMBER	COVER (mm)			EXPOSURE CLASSIF.
	TOP	BOTTOM	SIDES	
INTERNAL SLABS	30mm	30mm	30mm	A1
EXTERNAL SLABS	40mm	40mm	40mm	A2
PIERS	40mm	40mm	40mm	A2

CONCRETE QUALITY

ELEMENT	SLUMP	AGGREGATE (MAX. SIZE)	CEMENT TYPE	f 'c
SLAB ON GROUND	80mm	20mm	A	32MPa
PIERS	80mm	20mm	A	32MPa

MESH SCHEDULE

SLAB THICKNESS	BOTTOM REINFORCEMENT	TOP REINFORCEMENT
100	-	SL82 MESH



NORTH POINT

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GEMSTRUX CONSULTING ENGINEERS

CLIENT: DAVID BOURCHAN

SITE: 16 HENSON STREET, MERRYLANDS, NSW 2160

TITLE: GROUND FLOOR SLAB PLAN

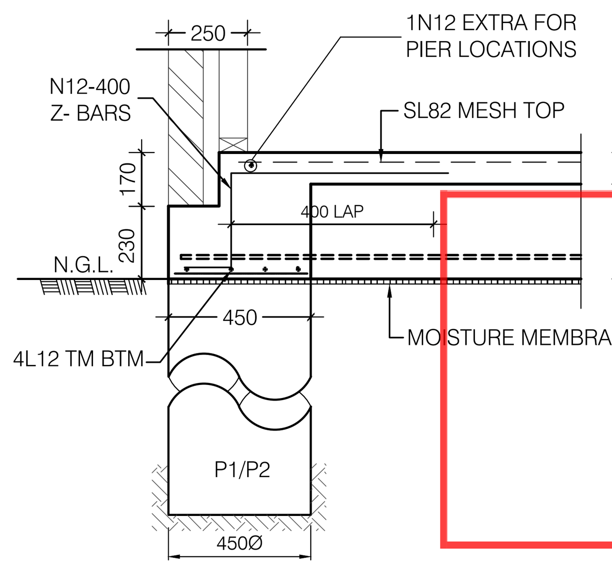
APPROVED: J.M. B.E CONSTRUCTION MIEAust MEMB. NO: 4468815 | DESIGNED: J.M. | DRAWN: J.P.F.

0 1 2 3 4 5 10
GRAPHIC SCALE: 1:100

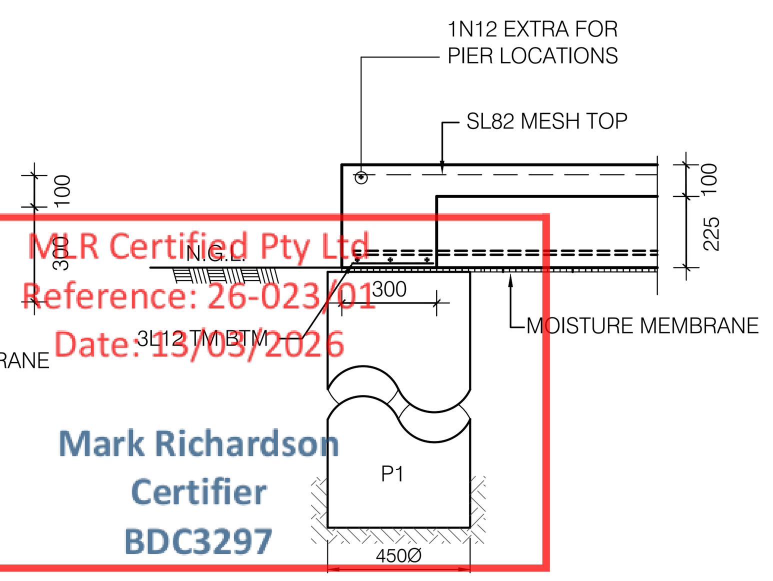
PROJECT NO: G26043

DRAWING NO: S6

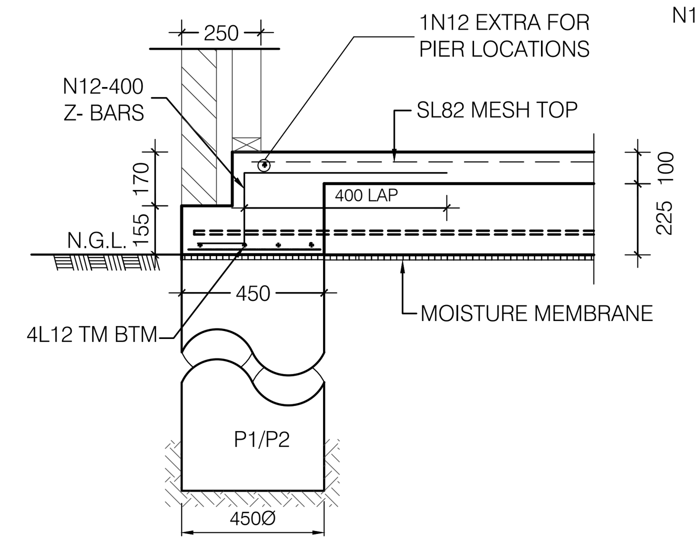
SCALE @ A1 1:100 U.N.O



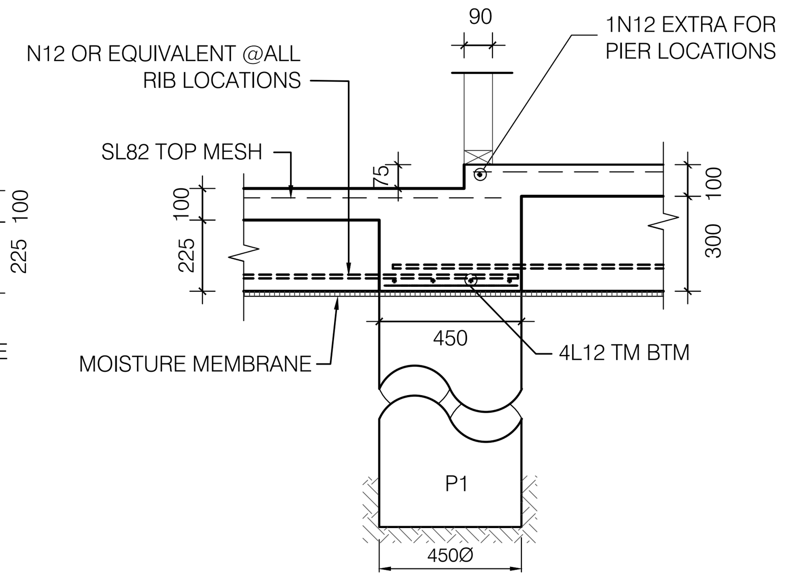
EB1 EDGE BEAM



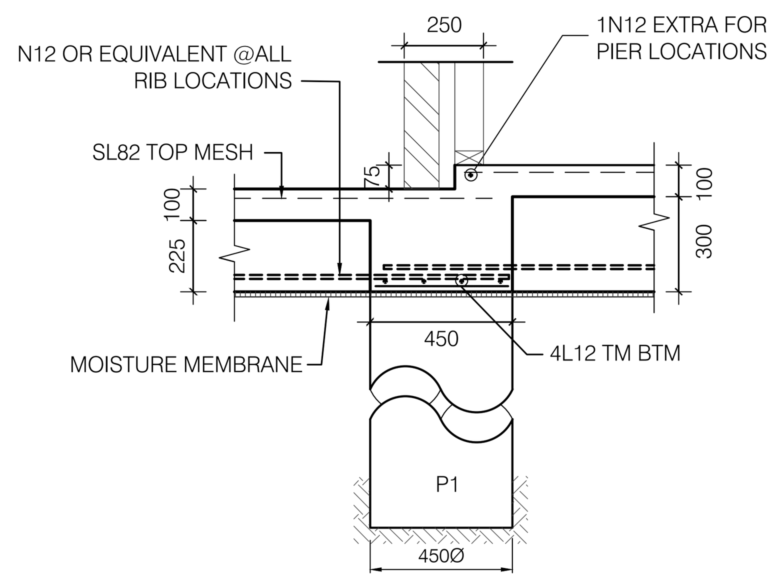
EB2 EDGE BEAM



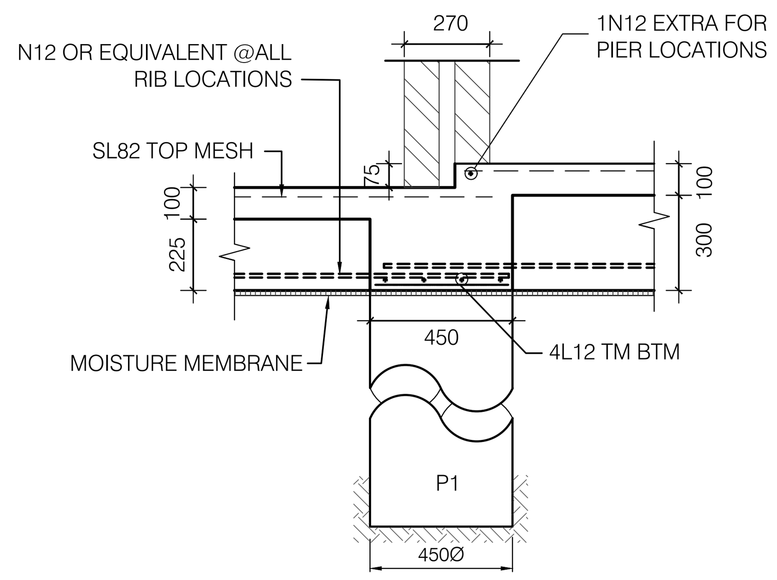
EB3 EDGE BEAM



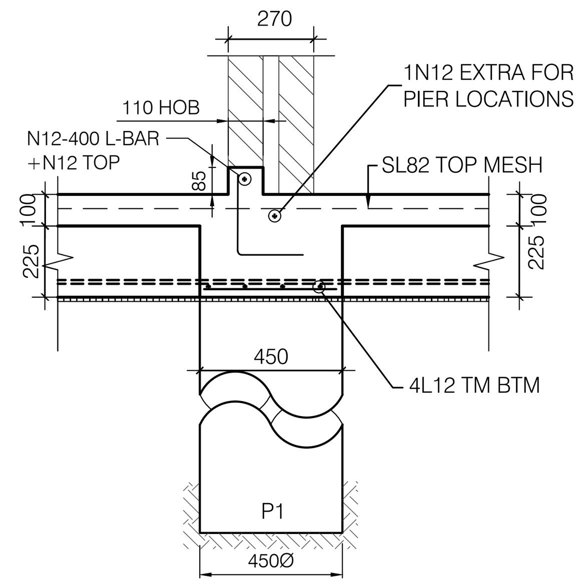
SB1 STEP BEAM



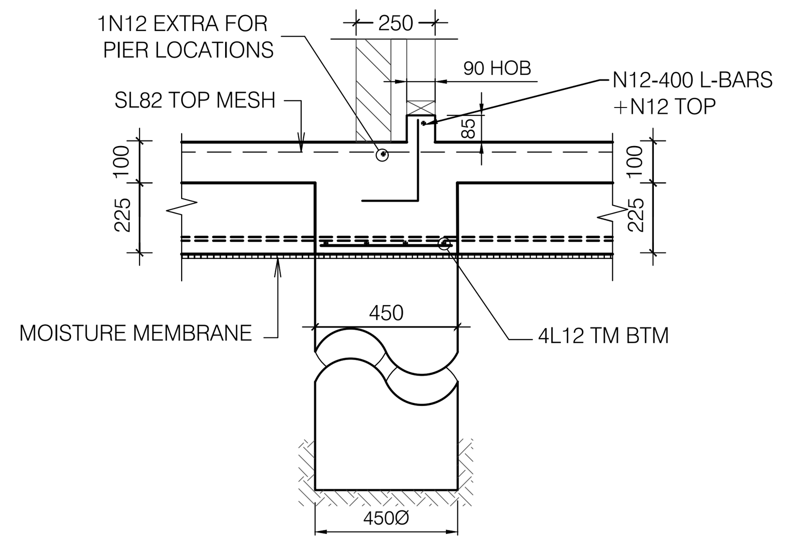
SB2 STEP BEAM



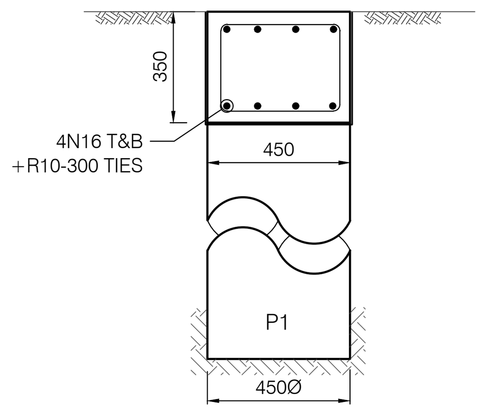
SB3 STEP BEAM



IB1 INTERNAL BEAM

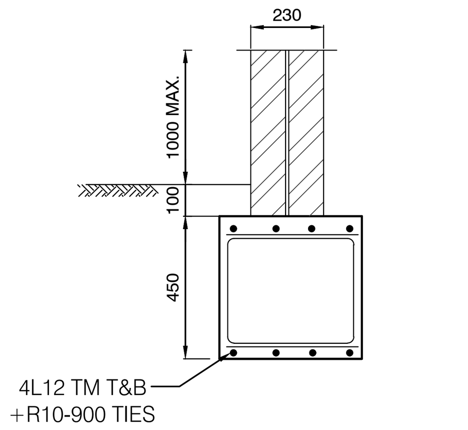


IB2 INTERNAL BEAM



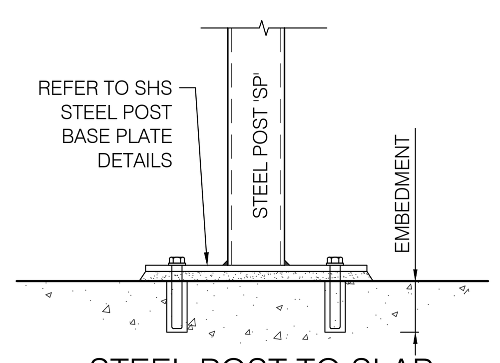
BRIDGE FOOTING - BF1

SCALE 1:20

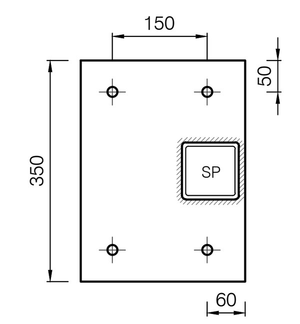


FF1 TYPICAL FENCE FOOTING DETAIL

SCALE 1:20

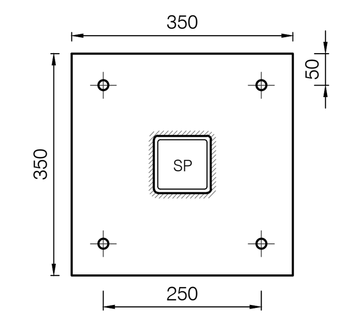


STEEL POST TO SLAB CONNECTION DETAIL



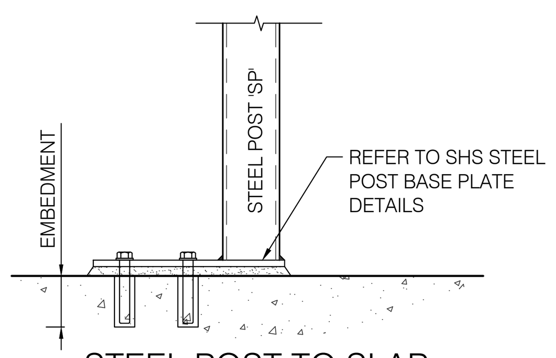
EDGE

1. 10mm PLATE (8mm CFW)
2. 4M16 CHEMSET ANCHOR BOLTS WITH 75mm EMBEDMENT
3. 15mm NON-SHRINK GROUT

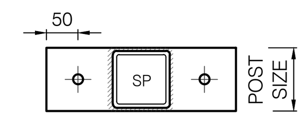


CENTER

1. 10mm PLATE (8mm CFW)
2. 4M16 CHEMSET ANCHOR BOLTS WITH 75mm EMBEDMENT
3. 15mm NON-SHRINK GROUT

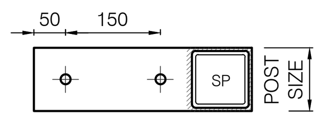


STEEL POST TO SLAB CONNECTION DETAIL



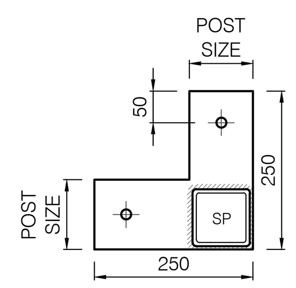
MIDDLE

1. 10mm PLATE (8mm CFW)
2. 2M16 CHEMSET ANCHOR BOLTS WITH 75mm EMBEDMENT
3. 15mm NON-SHRINK GROUT
4. ORIENTATE BASE PLATE TO SUIT WALL LOCATION



EDGE

1. 10mm PLATE (8mm CFW)
2. 2M16 CHEMSET ANCHOR BOLTS WITH 75mm EMBEDMENT
3. 15mm NON-SHRINK GROUT
4. ORIENTATE BASE PLATE TO SUIT WALL LOCATION



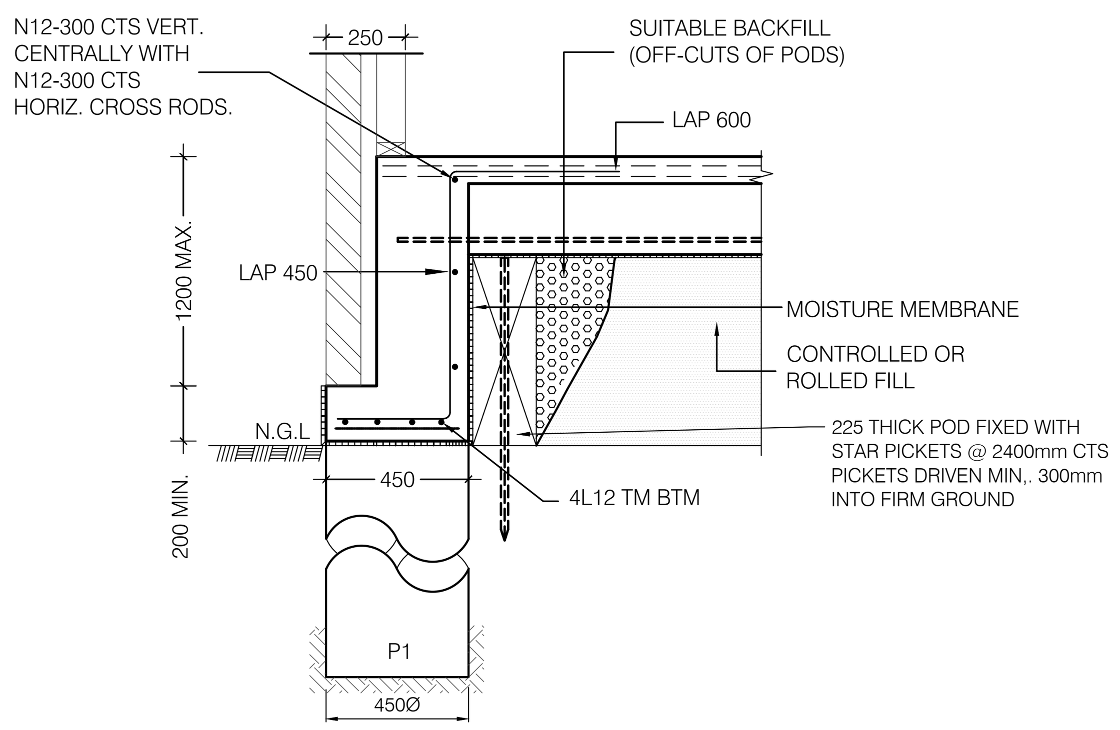
CORNER

1. 10mm PLATE (8mm CFW)
2. 2M16 CHEMSET ANCHOR BOLTS WITH 75mm EMBEDMENT
3. 15mm NON-SHRINK GROUT
4. ORIENTATE BASE PLATE TO SUIT WALL LOCATION

SHS STEEL POST BASE PLATE DETAILS

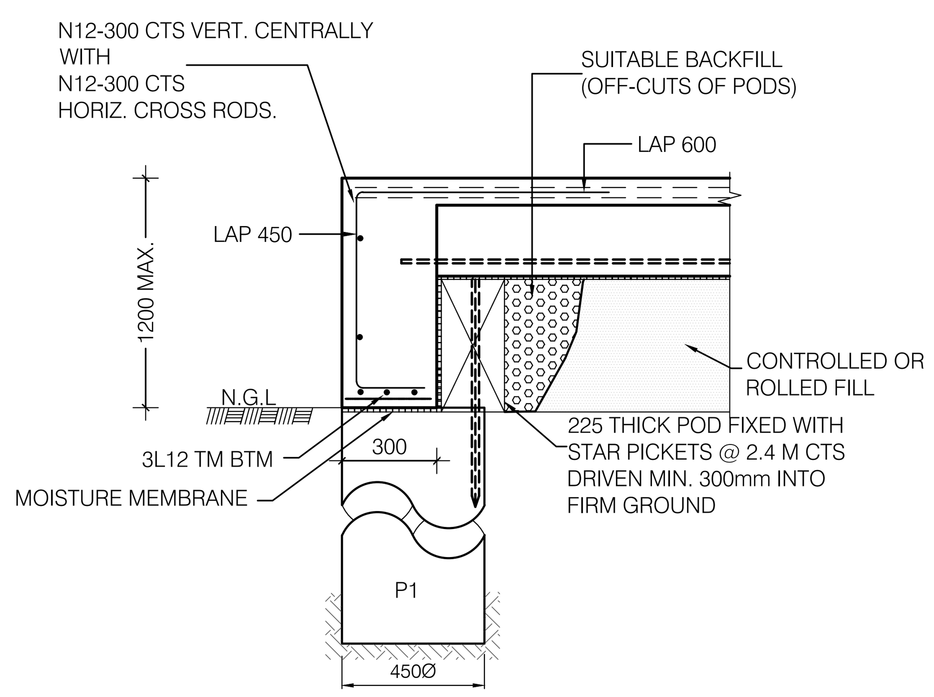
SCALE 1:10

SCALE 1:10



DEB1 DROPPED EDGE BEAM (1200 MAX.)

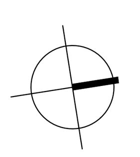
WHERE REQUIRED



DEB2 DROPPED EDGE BEAM (1200 MAX.)

WHERE REQUIRED

NORTH POINT



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CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
GROUND FLOOR
SECTIONS & DETAILS

APPROVED: J.M.
B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

0 0.1 0.2 0.3 0.4 0.5 1.0 1.5 2.0
GRAPHIC SCALE 1:20

PROJECT NO:
G26043

DRAWING NO:
S7

SCALE @ A1
1:20 U.N.O

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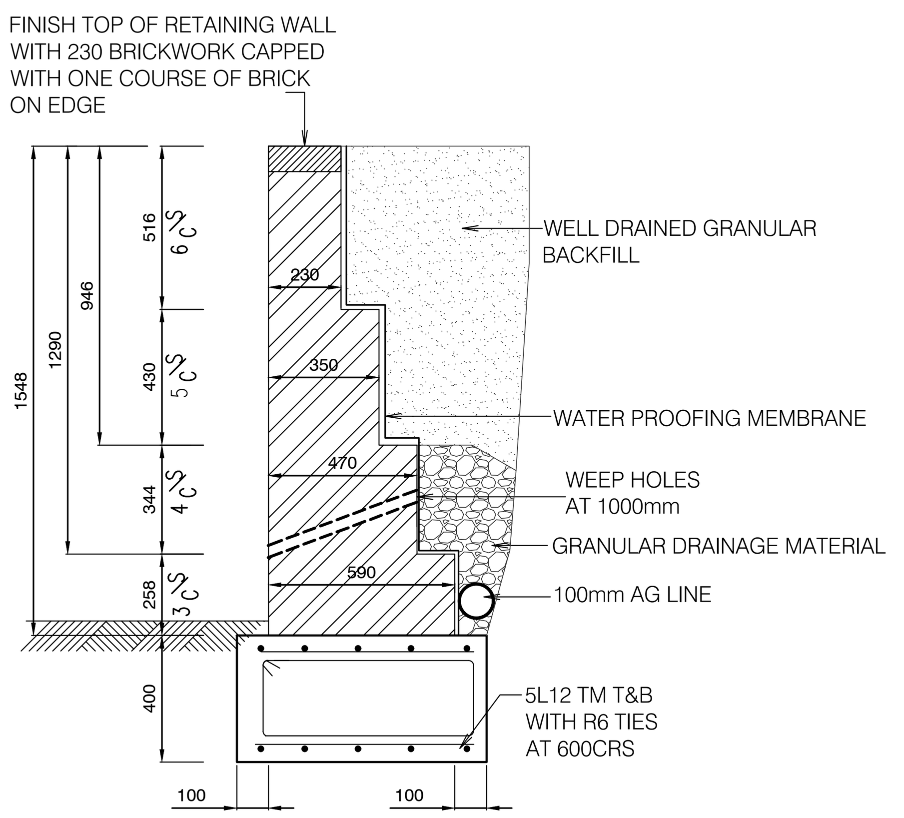
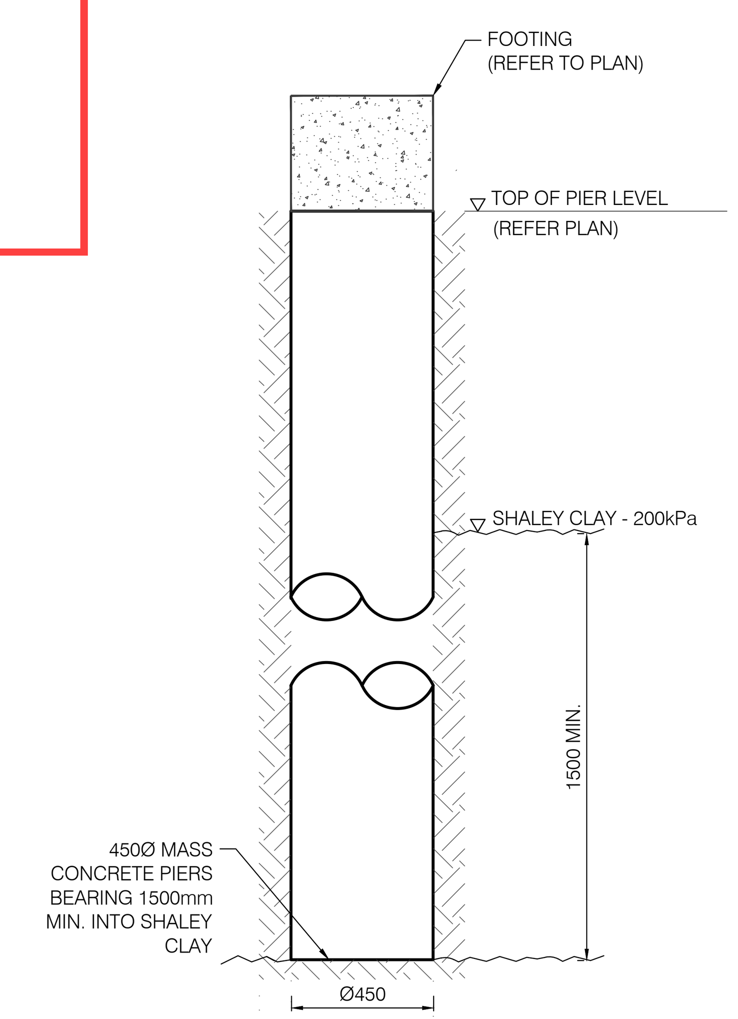
Reference: 26-023/01
Date: 13/03/2026

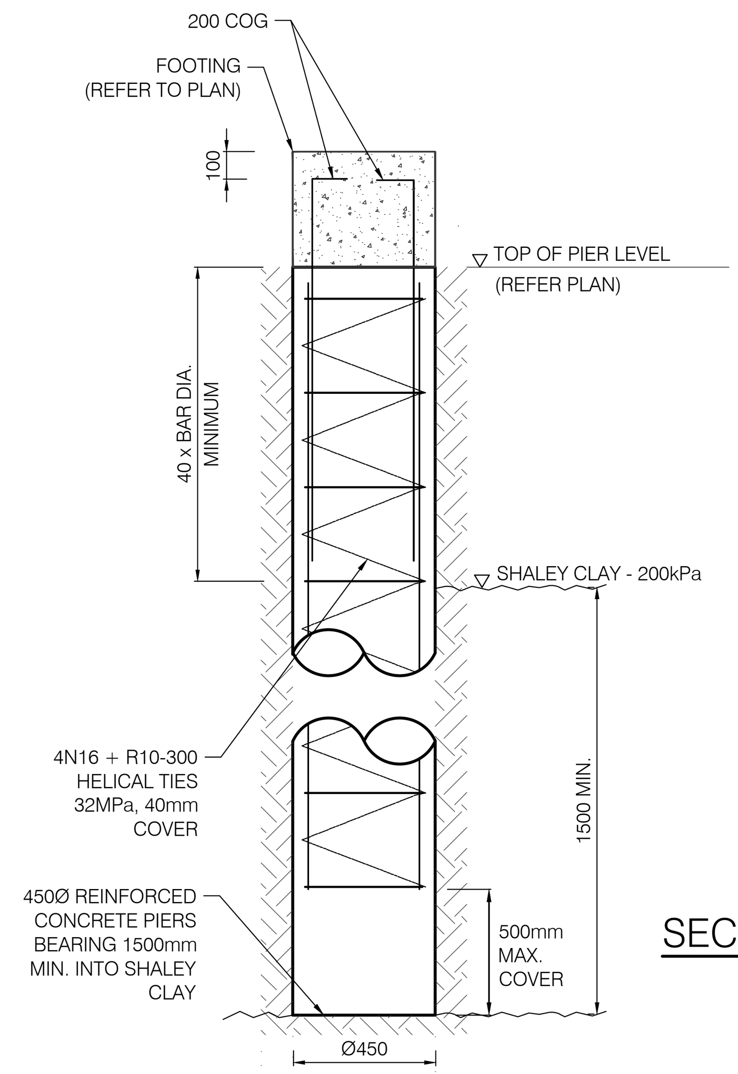
Mark Richardson
Certifier
BDC3297



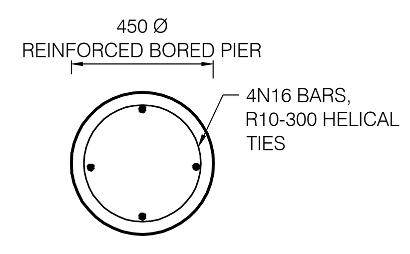
GARDEN RETAINING WALL



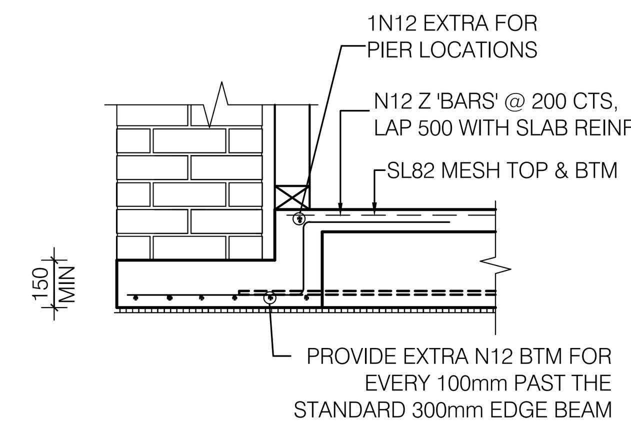
MASS CONCRETE BORED PIER DETAIL - (P1)
SCALE 1:20



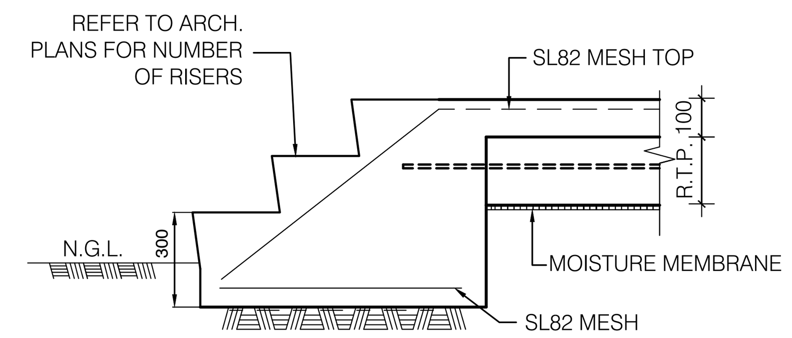
REINFORCED CONCRETE BORED PIER DETAIL - (P2)
SCALE 1:20



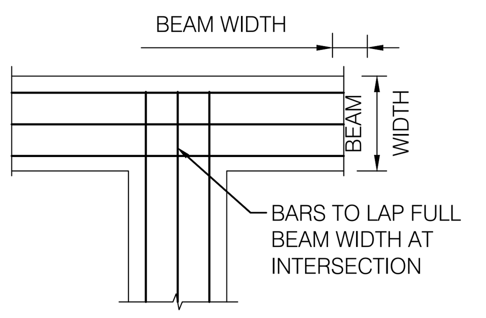
SECTION THROUGH PIER -P2
SCALE 1:20



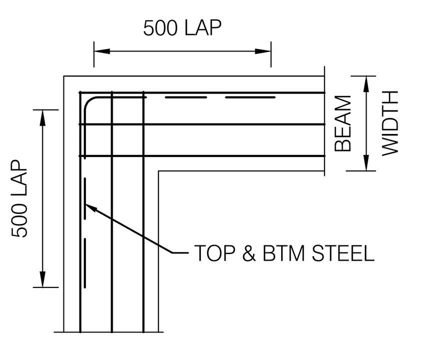
LOCAL WIDENING OF EDGE BEAM
APPLICABLE UNDER ENGAGED BRICK



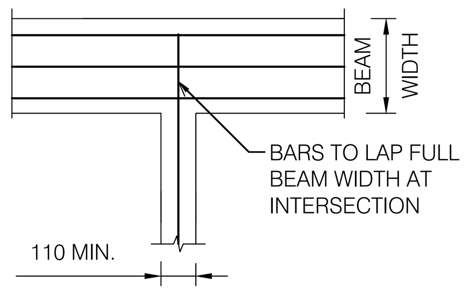
TYPICAL EXTERNAL STAIRS DETAIL



PLAN ON INTERSECTION

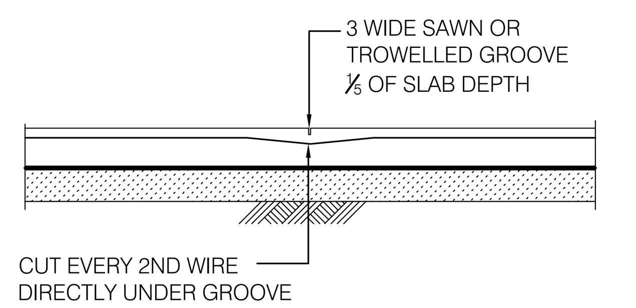


PLAN ON CORNER DETAIL

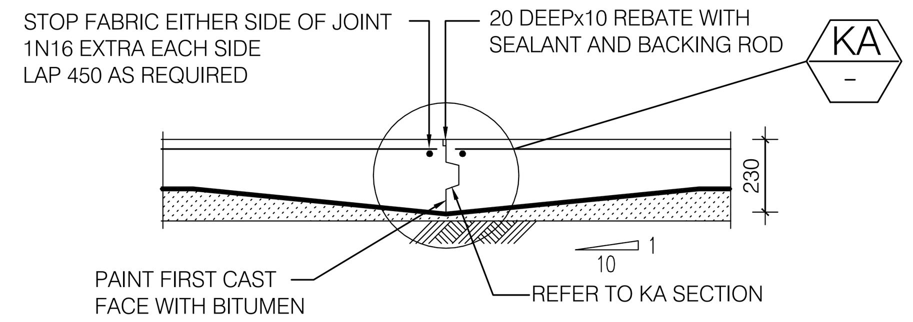


PLAN ON INTERSECTION

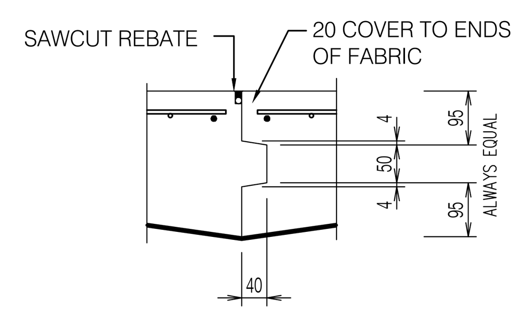
REINFORCING BARS SHALL HAVE A LAP LENGTH AT SPLICES OF 500mm, MIN. AT 'T' & 'L' INTERSECTIONS, THE BARS SHALL BE CONTINUES ACROSS THE FULL WIDTH OF THE RESPECTIVE INTERSECTIONS. AT L-INTERSECTINOS, A BENTBAR 500mm LONG ON EACH LEG SHALL BE PROVIDED. REFER TO DETAILS ABOVE.



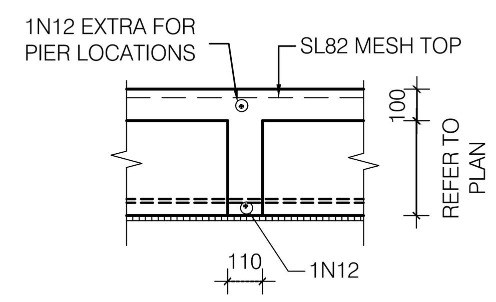
S.J. - SAWN CONTROL JOINT
SCALE 1:20
NOTE: SAWCUT TO BE MADE WITHIN 16 HOURS OF CASTING SLAB



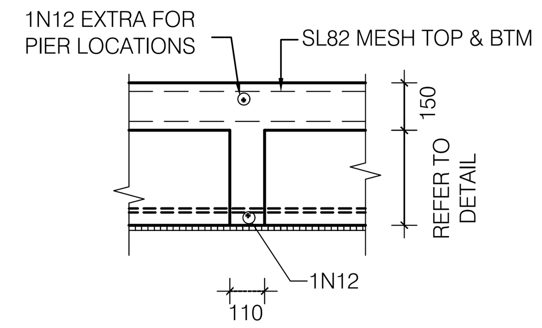
K.J. - KEYED CONSTRUCTION JOINT
SCALE 1:20



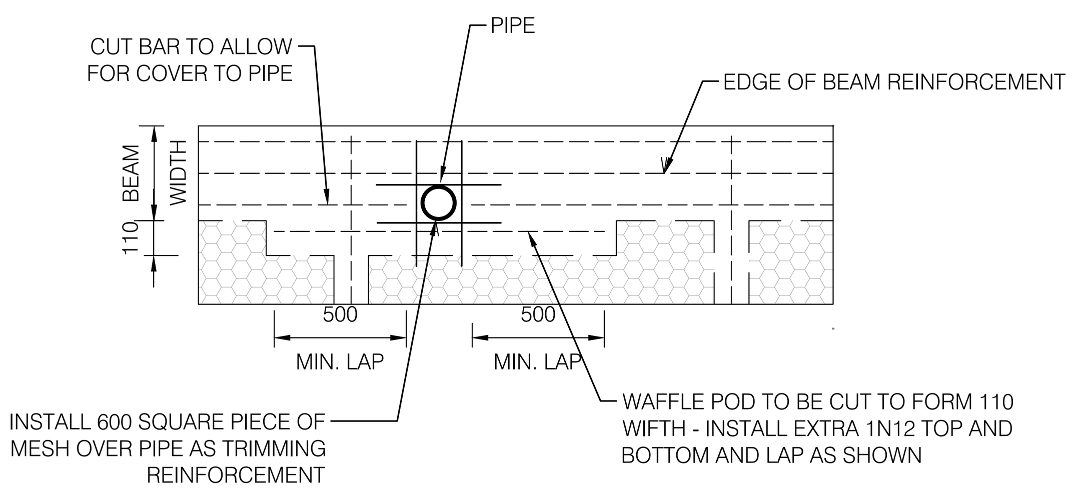
DETAIL KA
SCALE 1:10



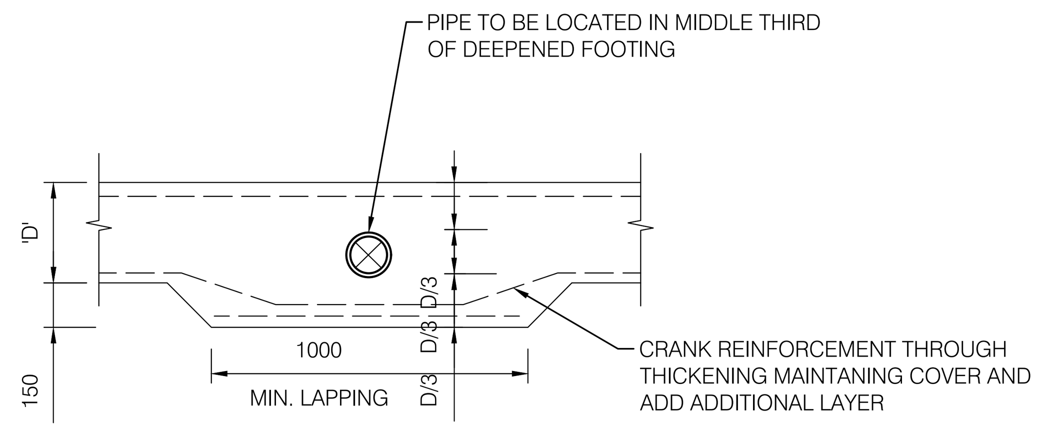
INTERNAL RIB - 100 THICK SLAB



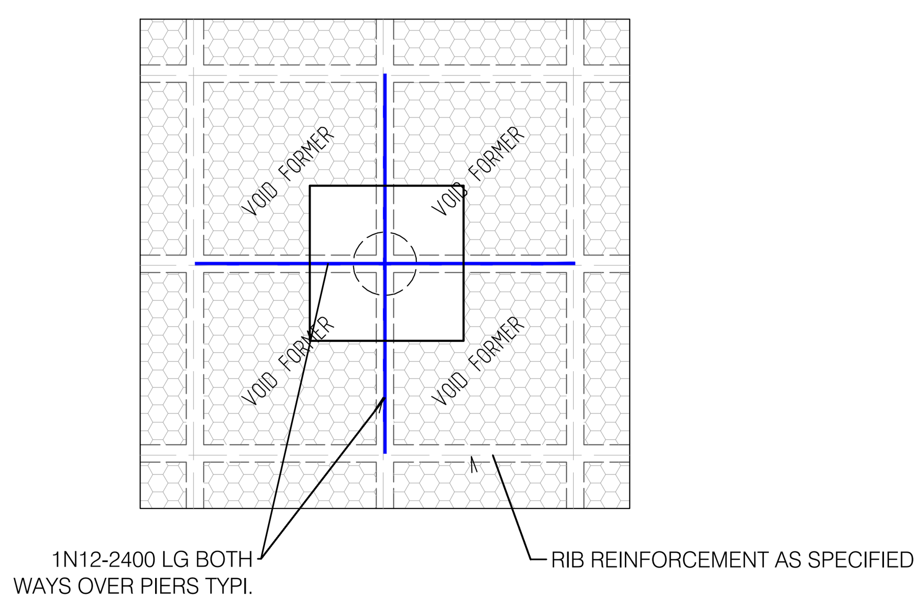
INTERNAL RIB- 150 THICK SLAB



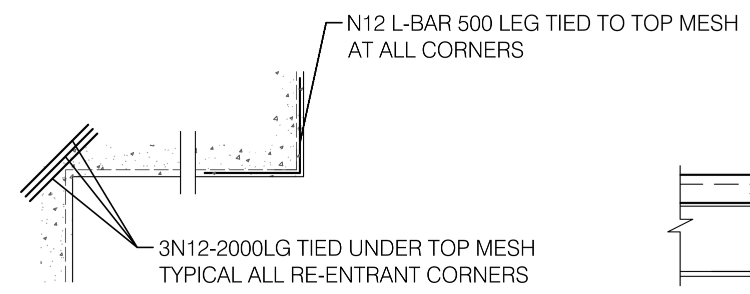
PLUMBING PIPE PENETRATION DETAILS
PLUMBING PIPES RO BE 100Ø MAX. WRAPPED WITH 20mm LAGGING AND FITTED WITH FLEXIBLE JOINTS TO ALLOW MOVEMENT OF UP TO 50mm IN ANY DIRECTION



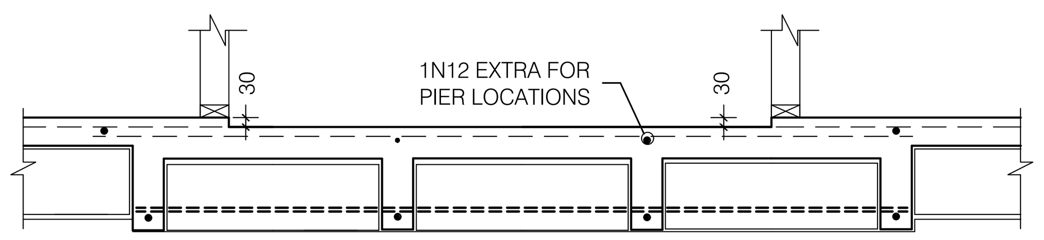
ELEVATIONS FOR HORIZONTAL PIPE THROUGH EDGE BEAM
SCALE 1:20



TYPICAL REINFORCEMENT FOR INTERNAL PIERS (WHERE REQUIRED)
SCALE 1:40



TRIMMER BARS DETAIL
SCALE 1:20



TYPICAL TILE (30mm) SETDOWN DETAIL

NORTH POINT

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REVISIONS

Rev	Date	Description	Verified
REV A	13/02/2026	ISSUED FOR COORDINATION	J.M.

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GEMSTRUX
CONSULTING ENGINEERS

CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
GROUND FLOOR
SECTIONS & DETAILS

APPROVED: J.M.
B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

0 0.1 0.2 0.3 0.4 0.5 1.0 1.5 2.0
GRAPHIC SCALE 1:20

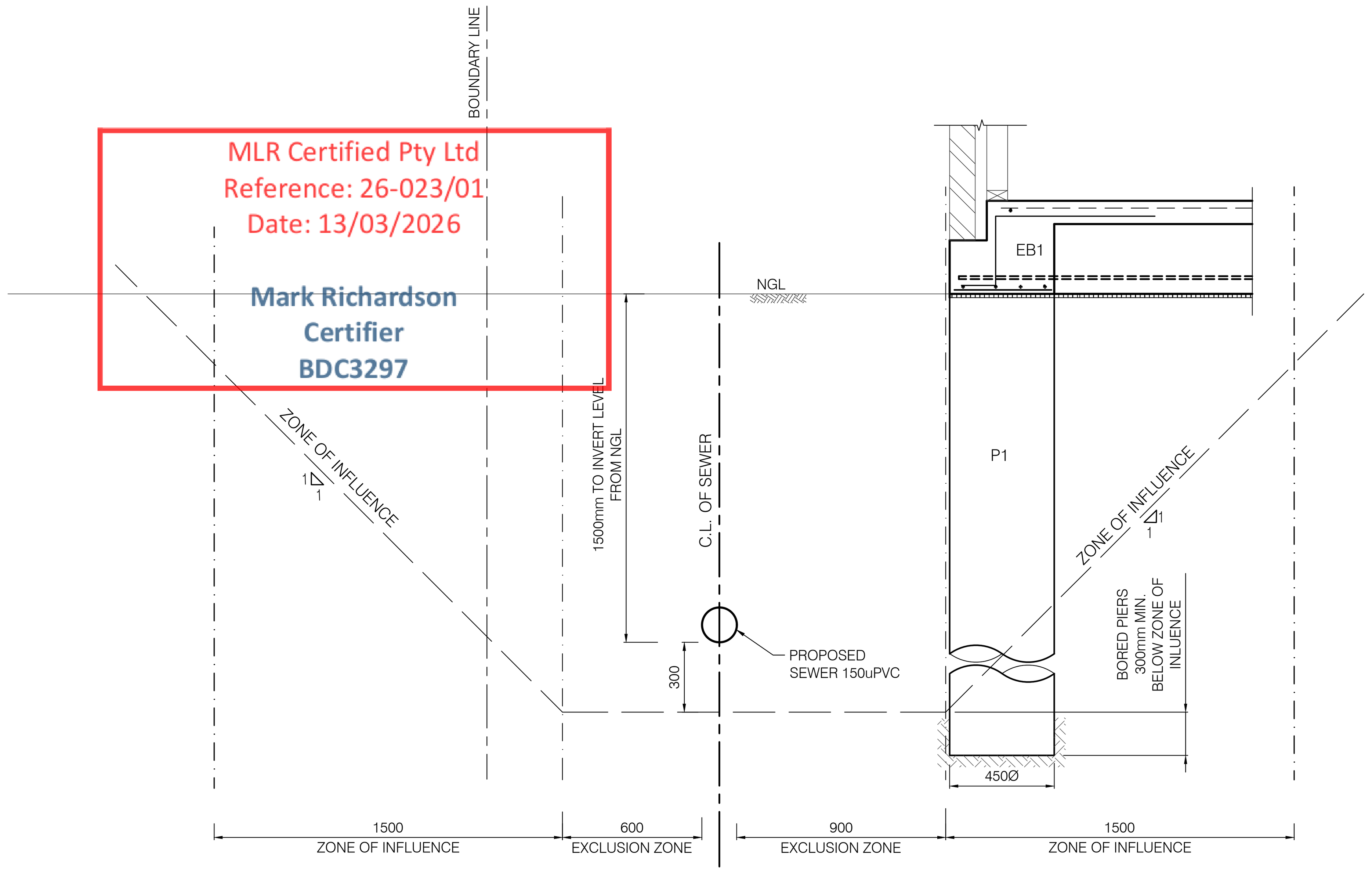
PROJECT NO:
G26043

DRAWING NO:
S8

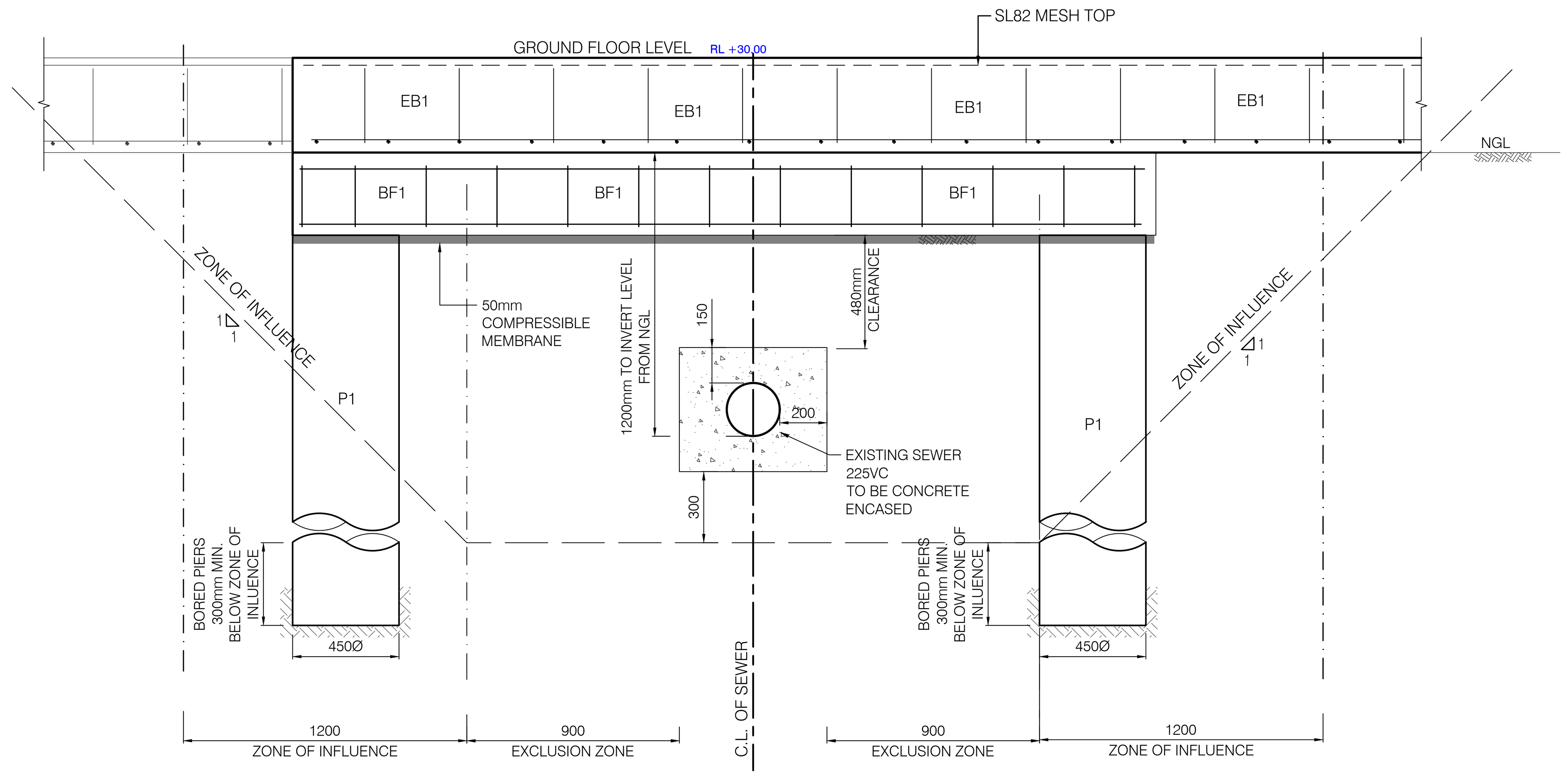
SCALE @ A1
1:20 U.N.O

MLR Certified Pty Ltd
Reference: 26-023/01
Date: 13/03/2026

Mark Richardson
Certifier
BDC3297



SECTION 1 S6
SCALE 1:20



SECTION 2 S6
SCALE 1:20

NORTH POINT

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GEMSTRUX
CONSULTING ENGINEERS

CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
SEWER SECTION

APPROVED: J.M.
B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

0 0.1 0.2 0.3 0.4 0.5 1.0 1.5 2.0
GRAPHIC SCALE 1:20

PROJECT NO:
G26043

DRAWING NO:
S9

SCALE @ A1
1:20 U.N.O

FIRST FLOOR AND LOWER ROOF FRAMING PLAN

NOTES:
1. ALL EXPOSED STEEL TO BE PRESSURE GALVANIZED
2. ALL STEEL FIXINGS TO BE IN ACCORDANCE TO AS4100
3. ALL EXPOSED TIMBER TO BE PRESSURE TREATED TO AS1684
4. ALL DETAILS TO BE CONFIRMED DURING CONSTRUCTION
5. ALL TIMBER TO BE IN ACCORDANCE TO AS1684
6. ROOF TO BE BUILT IN ACCORDANCE TO AS1684

MLR Certified Pty Ltd
Reference: 26-023/01
Date: 13/03/2026

Mark Richardson
Certifier
BDC3297

LEGEND

	BRICKWORK BELOW
	STUD WALL BELOW
	60mm WET AREA SETDOWN
	DENOTES VOID
	ROOF TO AS1684
TP/A	DENOTES 90x90 HARDWOOD POST ABOVE
TP/B	DENOTES 90x90 HARDWOOD POST BELOW
SP/B	DENOTES STEEL POST BELOW (89x89x6.0mm SHS) (HOT DIP GALV. (IF EXTERNAL OR EXPOSED)
	DIRECTION OF RAFTERS
BR1	25x1mm GALVANIZED STRAP - FIX TO EACH RAFTER WITH 2/60x3.15mm NAILS

TIMBER MEMBER SCHEDULE

MARK	TYPE & GRADE	SIZE
R1	TIMBER RAFTER	90 x 45 MGP10 @ 600 CTS MAX SPAN 1.8 m SINGLE SPAN
R2	TIMBER RAFTER	90 x 45 MGP10 @ 600 CTS MAX SPAN 2.8 m CONTINUOUS
J1	TIMBER JOIST	300 - 45 HYJOIST @ 450 CTS MAX SPAN 3.8 m
J2	TIMBER JOIST	240 - 45 HYJOIST @ 450 CTS MAX SPAN 2.4 m
J3	TIMBER JOIST	190 x 45 F7 @ 450 CTS TREATED PINE
J4	TIMBER JOIST	300 - 63 HYJOIST @ 450 CTS MAX SPAN 4.8 m
TB1	TIMBER BEAM	2/300 x 45 hySPAN LVL
TB2	TIMBER BEAM	1/200 x 45 hySPAN LVL
TB3	TIMBER BEAM	1/200 x 63 hySPAN LVL
TB4	TIMBER BEAM	2/290 x 45 F7 (TREATED PINE)
TB5	TIMBER BEAM	2/190 x 45 F7 (TREATED PINE)
TB6	TIMBER BEAM	1/290 x 45 F7 (TREATED PINE)
TB7	TIMBER BEAM	1/140 x 45 F7 (TREATED PINE)
TB8	TIMBER BEAM	2/300 x 63 hySPAN LVL
TB9	TIMBER BEAM	2/200 x 63 hySPAN LVL
TB10	TIMBER BEAM	1/300 x 63 hySPAN LVL
TB11	TIMBER BEAM	2/200 x 45 hySPAN LVL
TB12	TIMBER BEAM	2/140 x 45 F7 (TREATED PINE)
TB13	TIMBER BEAM	2/240 x 45 hySPAN LVL
TB14	TIMBER BEAM	1/190 x 45 F7 (TREATED PINE)
N.B	NAILING BEAM	TO SUIT SIZE OF ADJOINING MEMBERS

STEEL MEMBER SCHEDULE

MARK	TYPE & GRADE	SIZE
SB1	STEEL BEAM - G300	310 UB 46.2
SB2	STEEL BEAM - G300	310 UC 96.8
SB3	STEEL BEAM - G300	310 UC 158
SB4	STEEL BEAM - G300	300 PFC
SB5	STEEL BEAM - G300	250 PFC
SL1	STEEL LINTEL - G300	150 x 100 x 10mm UNEQUAL ANGLE
SL2	STEEL LINTEL - G300	100 x 100 x 10mm EQUAL ANGLE
P1	STEEL PLATE - G300	240 (WIDE) 20mm (THICK) PLATE
SP1	STEEL POST	89 x 89 x 6mm SHS

BRICK LINTEL SCHEDULE

OPENING (mm)	INTERNAL SKIN	EXTERNAL SKIN	END BEARING
<900	100 x 8mm FLAT BAR	100 x 6mm FLAT BAR	100mm
1200	100 x 10mm FLAT BAR	100 x 8mm FLAT BAR	100mm
1500	100 x 100 x 8mm ANGLE	100 x 100 x 6mm ANGLE	150mm
2100	150 x 100 x 8mm ANGLE	150 x 100 x 6mm ANGLE	150mm
2400	150 x 100 x 8mm ANGLE	150 x 100 x 10mm ANGLE	150mm
2700	150 x 100 x 10mm ANGLE	150 x 100 x 10mm ANGLE	150mm
3000	150 x 100 x 12mm ANGLE	150 x 100 x 12mm ANGLE	150mm

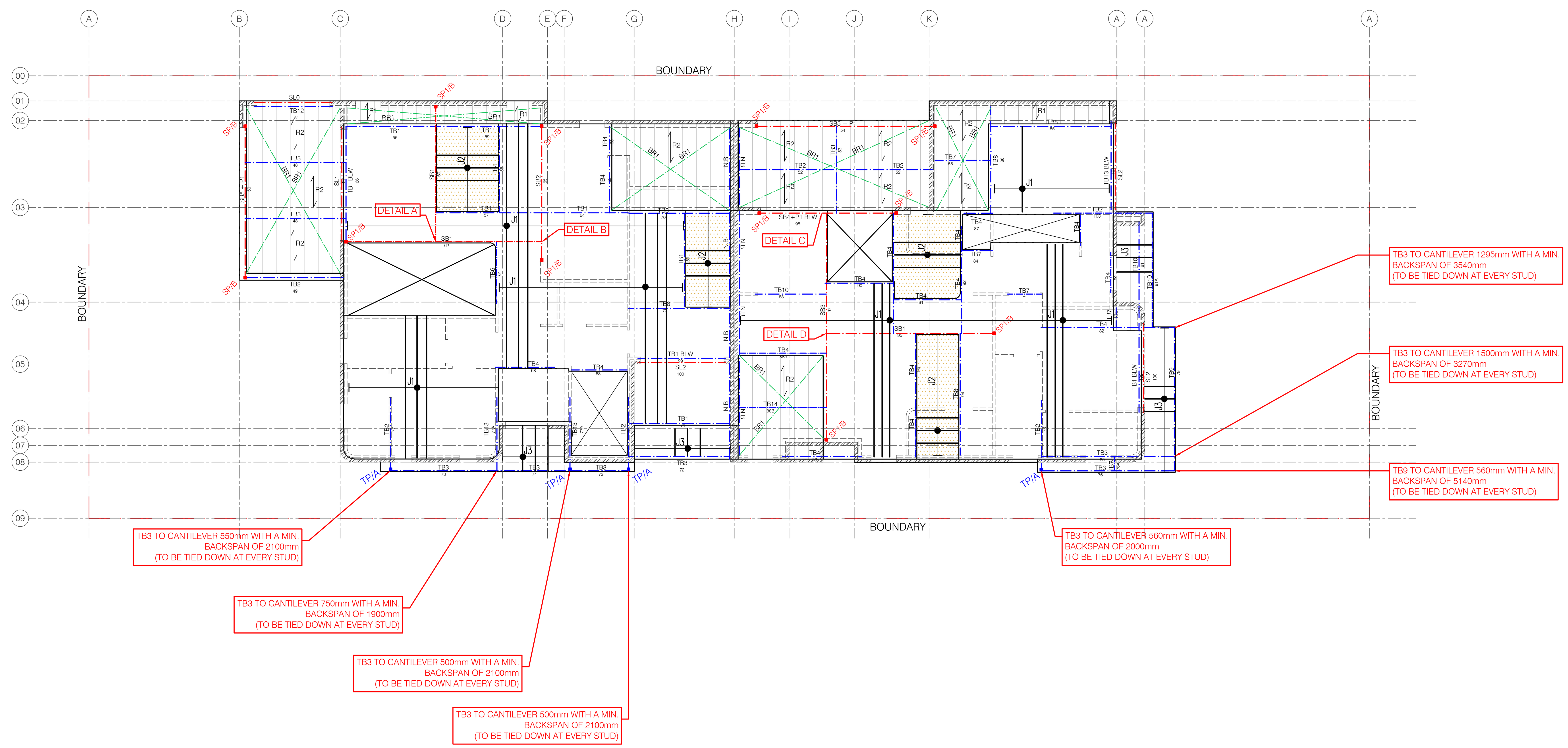
ALL EXPOSED STEEL MEMBERS TO BE HOT DIP GALVANISED

ALL BEAMS TO BE SUPPORTED BY DOUBLE STUDS IF NO POST IS DENOTED ON PLAN

ALL WALLS TO BE SUPPORTED BY DOUBLE JOISTS U.N.O.

ALL EXPOSED TIMBER TO BE TREATED OR H3

ALL STEEL BEAMS TO HAVE STIFFENER PLATES WELDED AT SUPPORTS



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GEMSTRUX CONSULTING ENGINEERS

CLIENT: DAVID BOURCHAN

SITE: 16 HENSON STREET, MERRYLANDS, NSW 2160

TITLE: FIRST FLOOR AND LOWER ROOF FARMING PLAN

PROJECT NO: G26043

APPROVED: J.M. B.E CONSTRUCTION MIEAust MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

DRAWING NO: S10

0 1 2 3 4 5 10
GRAPHIC SCALE: 1:100

SCALE @ A1: 1:100 U.N.O

WALL BRACING PLAN

NOTES:
1. WALL STRAP BRACING REFER TO DETAIL
2. STUD WALLS, NOGGINGS, TOP & BOTTOM PLATES TO BE MGP10 90x45

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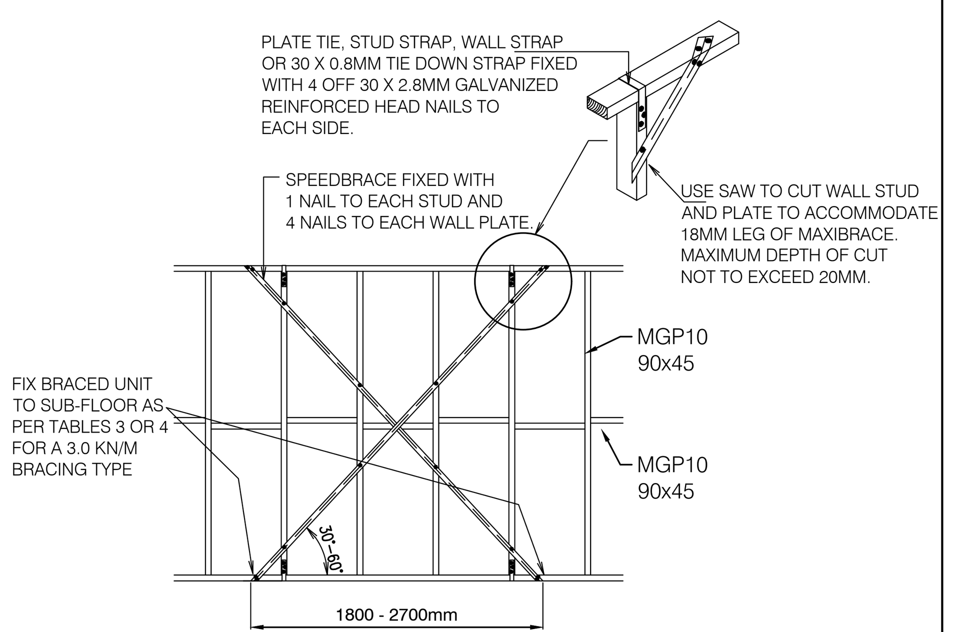
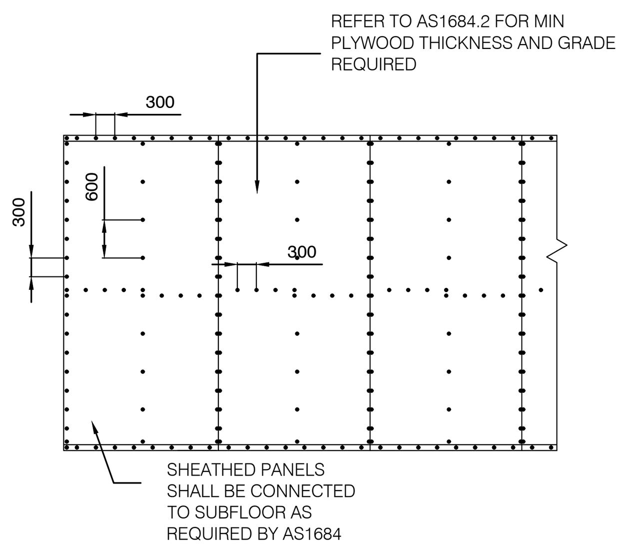
Reference: 26-023/01
Date: 13/03/2026
Mark Richardson
Certifier
BDC3297

LEGEND	
BR1	WALL STRAP SPEEDBRACE BRACING TYPE (REFER TO DETAIL)
BR2	WALL STRAP PLYWOOD BRACING TYPE (REFER TO DETAIL)
	BRICKWORK
	STUD WALL

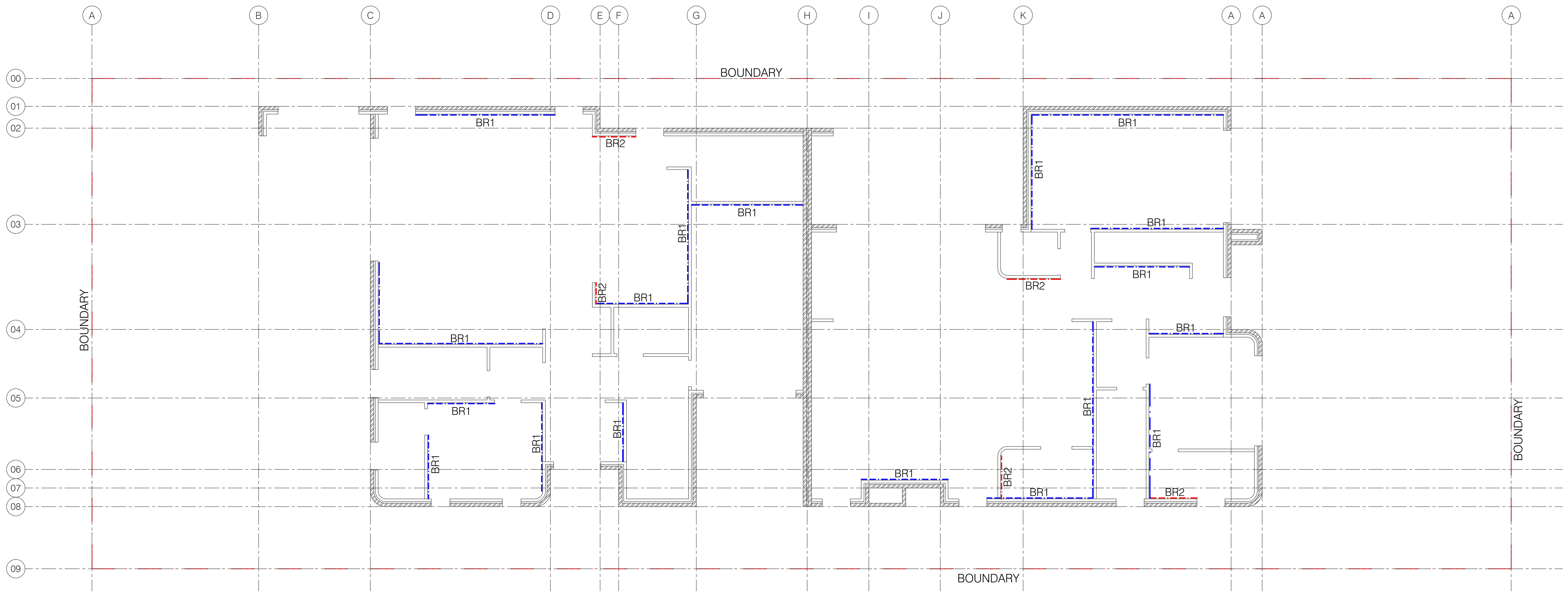
WIND LOADS	
REGION	A2
TERRAIN CATEGORY	3.0
GUST WIND SPEED V (m/s)	45
SHIELDING	1.0
TOPOGRAPIC	1.0
IMPORTANCE	2.0



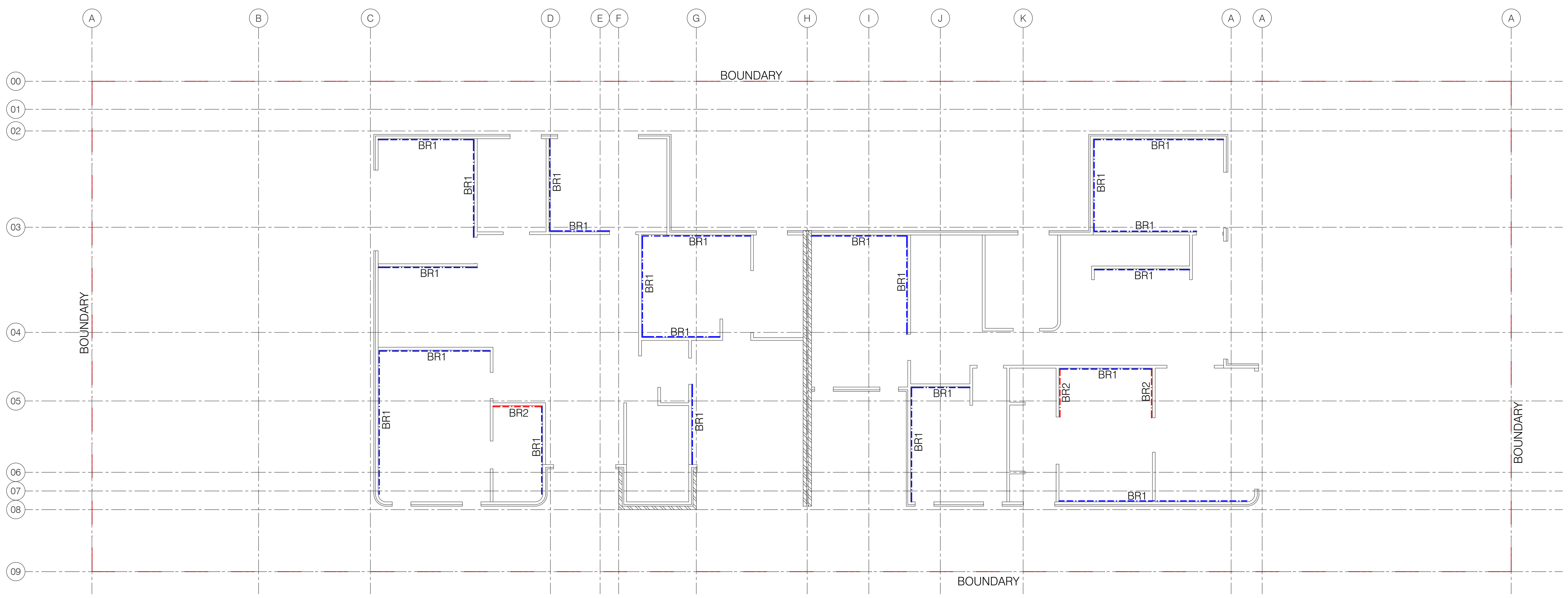
MAXI SPEEDBRACE
3.0kN/m BRACING TYPE
-BR1-



PLYWOOD BRACING
3.4kN/m BRACING TYPE
-BR2-

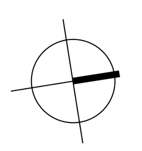


GROUND FLOOR WALLS
SCALE 1:100



FIRST FLOOR WALLS
SCALE 1:100

NORTH POINT



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CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
WALL BRACING PLANS

PROJECT NO:
G26043

APPROVED: J.M.
B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

DESIGNED: J.M.

DRAWN: J.P.F.

DRAWING NO:
S11

GRAPHIC SCALE: 1:100

SCALE @ A1
1:100 U.N.O

ROOF FRAMING PLAN

NOTES:
1. ALL STEEL FIXINGS TO BE IN ACCORDANCE TO AS4100
2. ALL EXPOSED TIMBER TO BE PRESSURE TREATED TO AS1684
3. ALL DETAILS TO BE CONFIRMED DURING CONSTRUCTION
4. ALL TIMBER TO BE IN ACCORDANCE TO AS1684
5. ROOF TO BE BUILT IN ACCORDANCE TO AS1684

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Reference: 26-023/01
Date: 13/03/2026

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Certifier
BDC3297

LEGEND

	STUD WALL BELOW
	ROOF TO AS1684
	DENOTES DIRECTION OF RAFTERS
BR1	25x1mm GALVANIZED STRAP - FIX TO EACH RAFTER WITH 2/60x3.15mm NAILS
	DENOTES VOID
TP/B	DENOTES 90x90 HARDWOOD POST BELOW

TIMBER MEMBER SCHEDULE

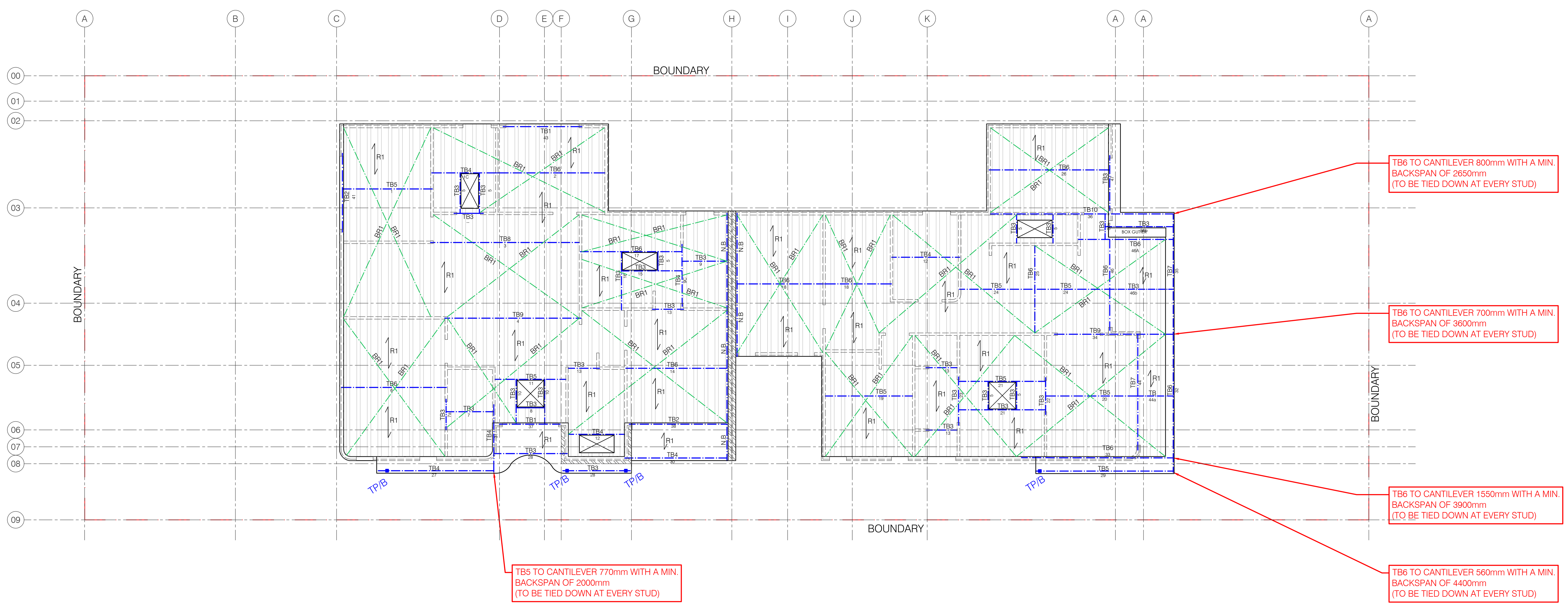
MARK	TYPE	SIZE
R1	TIMBER RAFTER	90 x 45 MGP10 @ 600 CTS MAX SPAN 2.8m CONTINUOUS
TB1	TIMBER BEAM	2/140 x 45 F7 (TREATED PINE)
TB2	TIMBER BEAM	2/190 x 45 F7 (TREATED PINE)
TB3	TIMBER BEAM	1/140 x 45 F7 (TREATED PINE)
TB4	TIMBER BEAM	1/190 x 45 F7 (TREATED PINE)
TB5	TIMBER BEAM	1/200 x 45 hySPAN LVL
TB6	TIMBER BEAM	1/200 x 63 hySPAN LVL
TB7	TIMBER BEAM	2/200 x 45 hySPAN LVL
TB8	TIMBER BEAM	2/240 x 45 hySPAN LVL
TB9	TIMBER BEAM	2/200 x 63 hySPAN LVL
TB10	TIMBER BEAM	2/300 x 63 hySPAN LVL
N.B	NAILING BEAM	TO SUIT SIZE OF ADJOINING MEMBERS

ALL EXPOSED STEEL MEMBERS TO BE HOT DIP GALVANISED

ALL BEAMS TO BE SUPPORTED BY DOUBLE STUDS IF NO POST IS DENOTED ON PLAN

ALL WALLS TO BE SUPPORTED BY DOUBLE JOISTS U.N.O.

ALL EXPOSED TIMBER TO BE TREATED OR H3



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CLIENT: DAVID BOURCHAN

SITE: 16 HENSON STREET, MERRYLANDS, NSW 2160

TITLE: ROOF FRAMING PLAN

PROJECT NO: G26043

APPROVED: J.M.
B.E CONSTRUCTION
MIEAust
MEMB. NO: 4468815

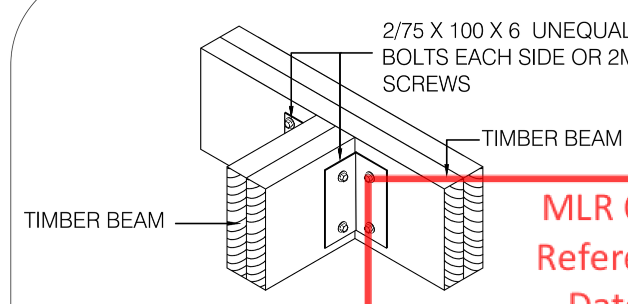
DESIGNED: J.M.

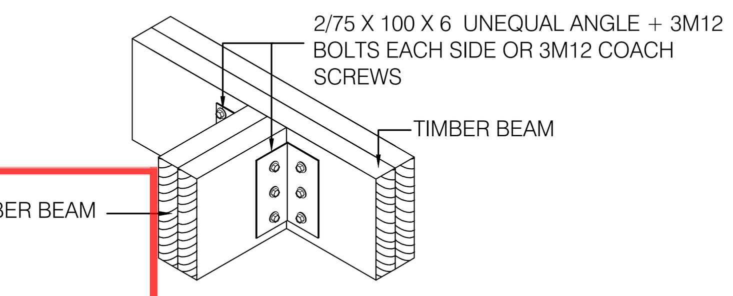
DRAWN: J.P.F.

DRAWING NO:

0 1 2 3 4 5 10
GRAPHIC SCALE: 1:100

SCALE @ A1
1:100 U.N.O





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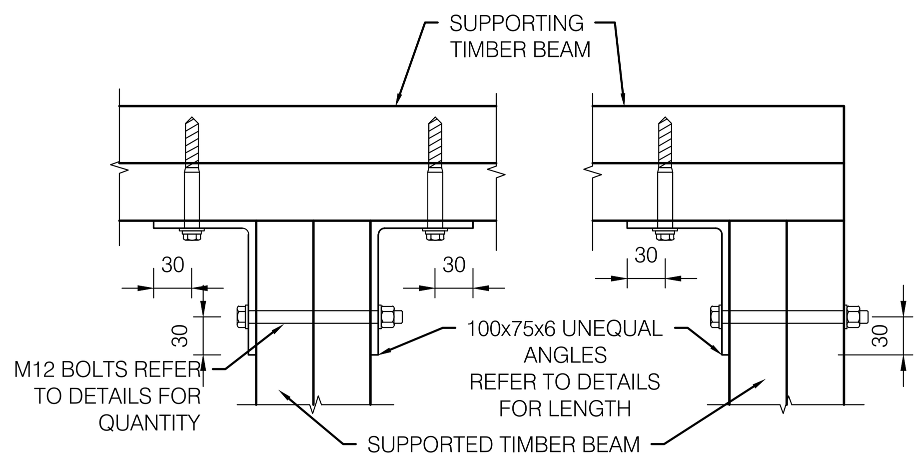
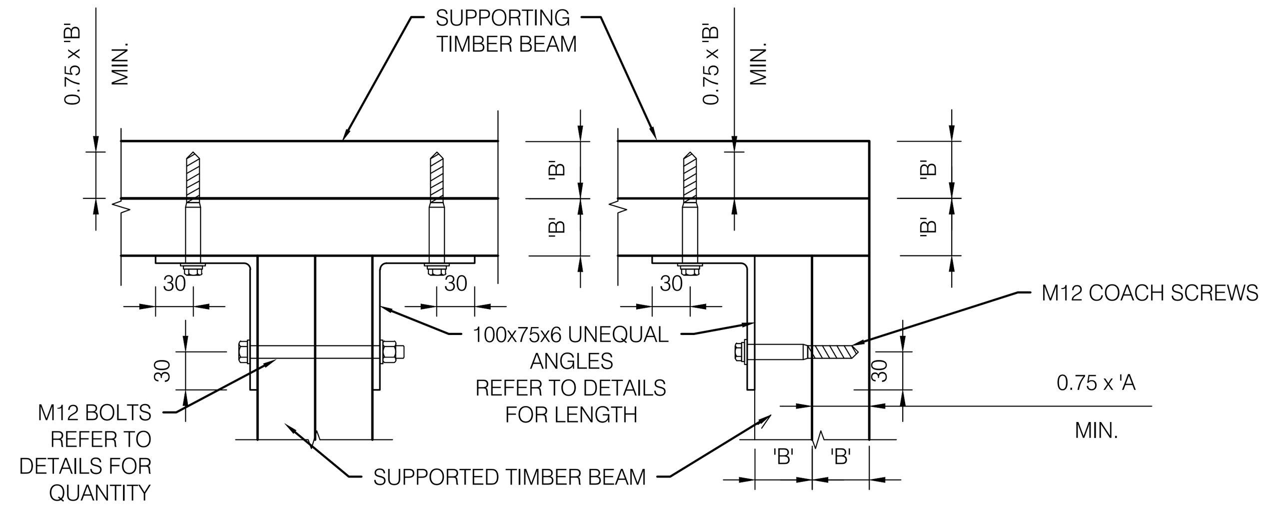
TIMBER BEAM CONNECTION
FOR DEPTHS LESSER THAN <200

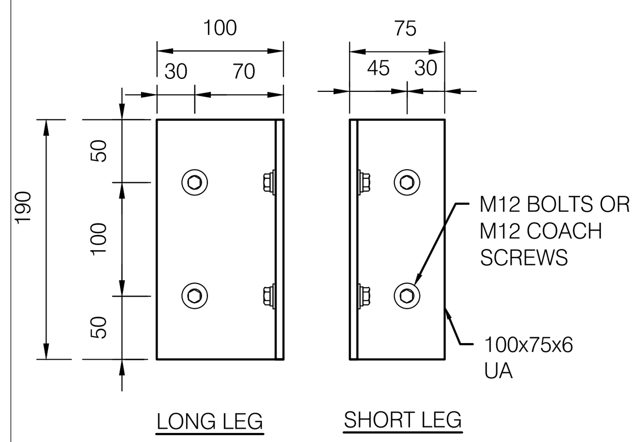
TIMBER BEAM CONNECTION
FOR DEPTHS GREATER THAN >200



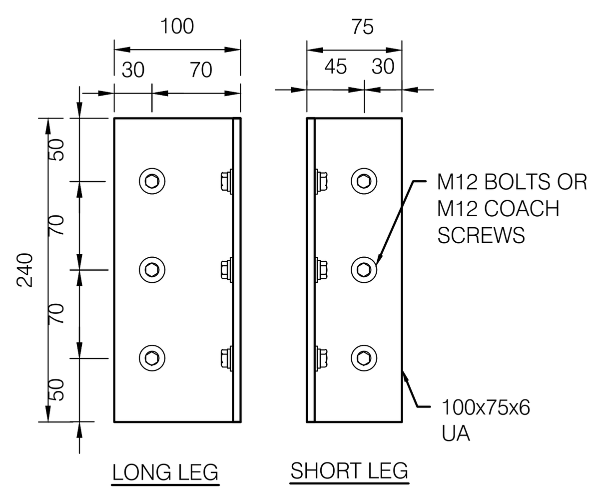
PLAN VIEW OF TYPICAL TIMBER BEAM CONNECTIONS
SCALE NTS



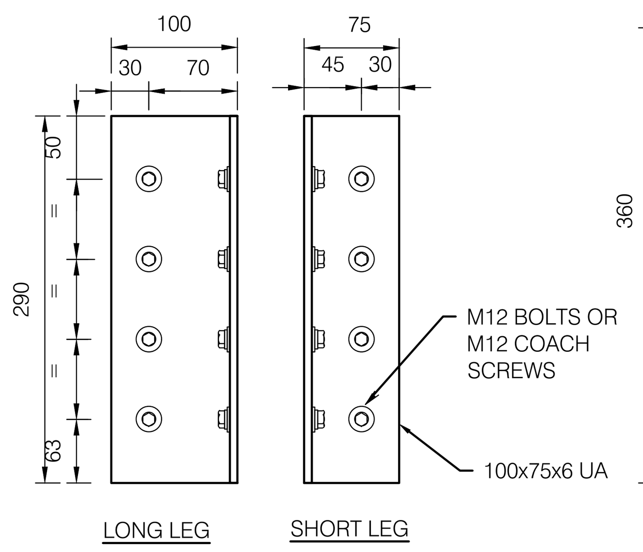
PLAN VIEW OF ALTERNATIVE TIMBER BEAM CONNECTIONS
SCALE NTS



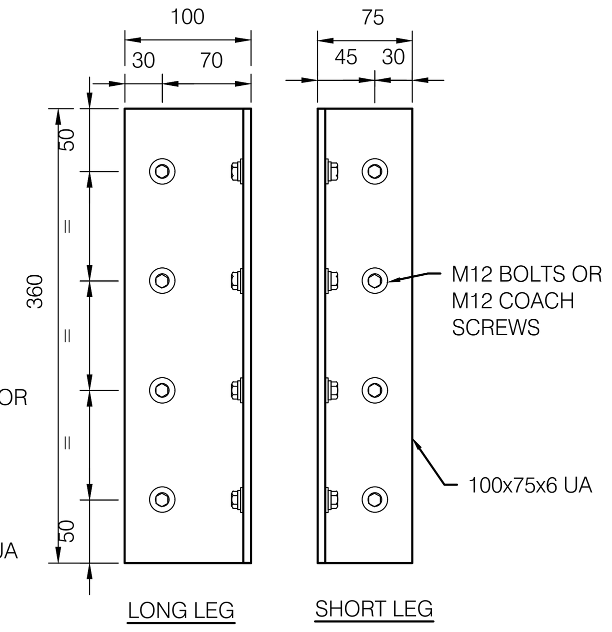
ANGLE TO SUIT SUPPORTED TIMBER BEAMS LESS THAN <200 DEEP



ANGLE TO SUIT SUPPORTED TIMBER BEAMS GREATER THAN >240 DEEP

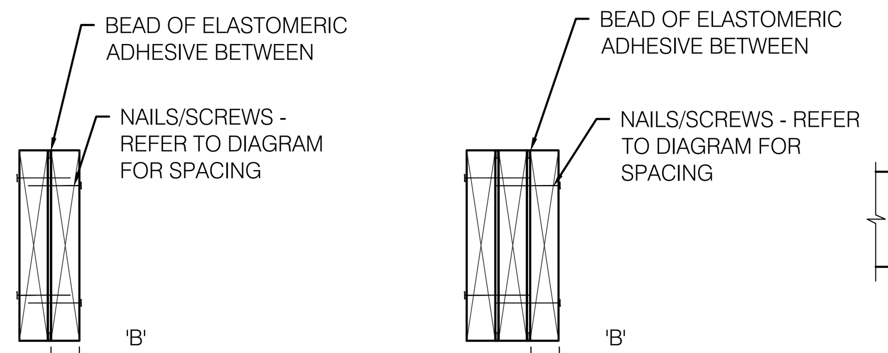


ANGLE TO SUIT 290/300 DEEP SUPPORTED TIMBER BEAMS



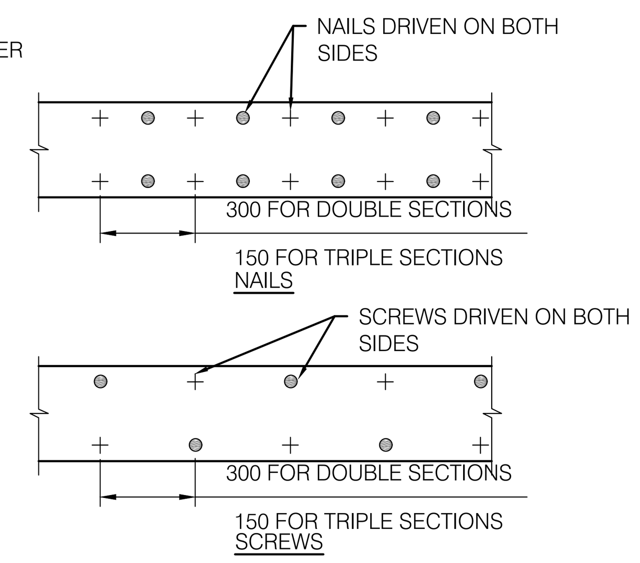
ANGLE TO SUIT 360 DEEP SUPPORTED TIMBER BEAMS

TIMBER BEAM CONNECTION DETAILS
SCALE NTS
NOTES:
1. ALL M12 BOLTS & COACH SCREWS TO BE FITTED WITH 55Ø x 3mm WAHERS WITH A SNUG FIT TO BOLT

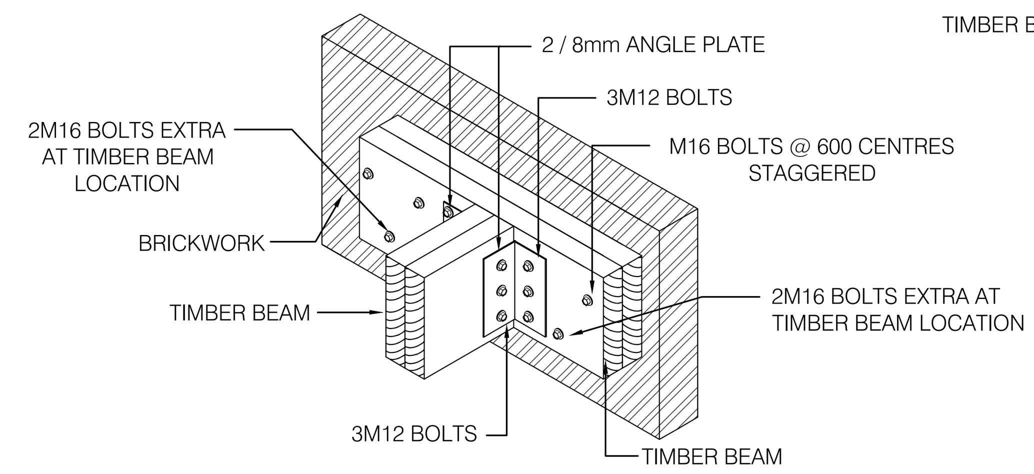


NAILING GUIDE

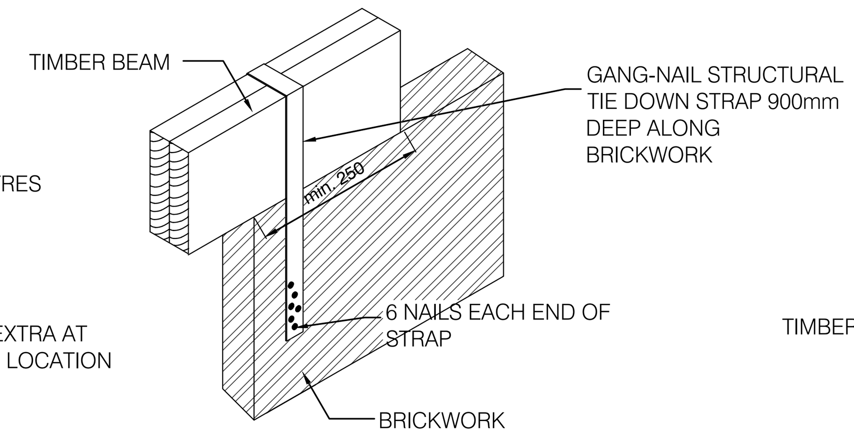
SECTION SIZE 'B'	MIN. NAIL DIA.	MIN. NAIL LENGTH	MIN.SCREW GUAGE	MIN.SCREW LENGTH
35	3.06mm	75mm	-	
45	3.30mm	90mm	14g	75mm
63	3.30mm	100mm	14g	100mm
90	-	-	14g	150mm



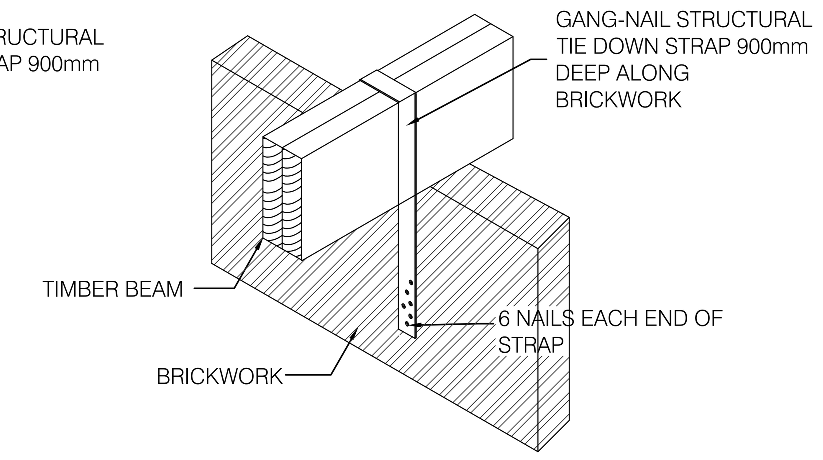
TIMBER BEAM LAMINATION DETAIL
SCALE NTS



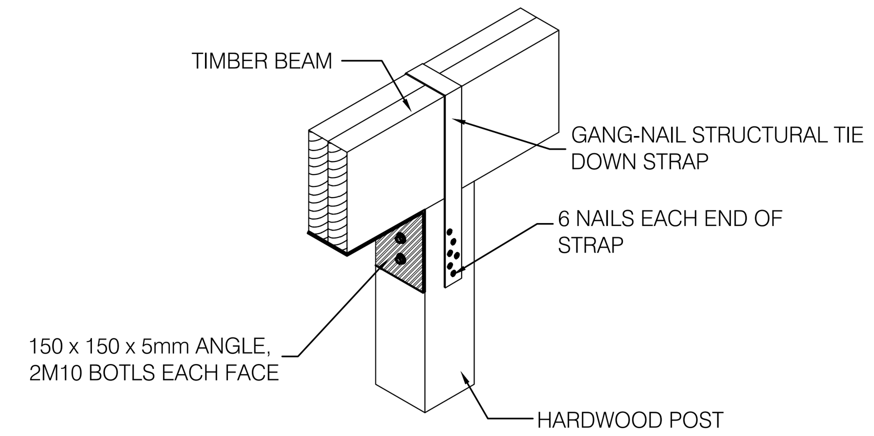
TIMBER BEAM TO BRICK WALL CONNECTION



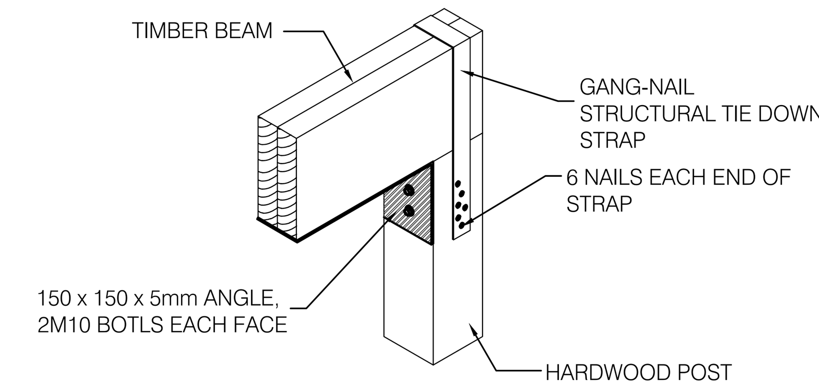
TIMBER BEAM TO BRICKWORK CONNECTION



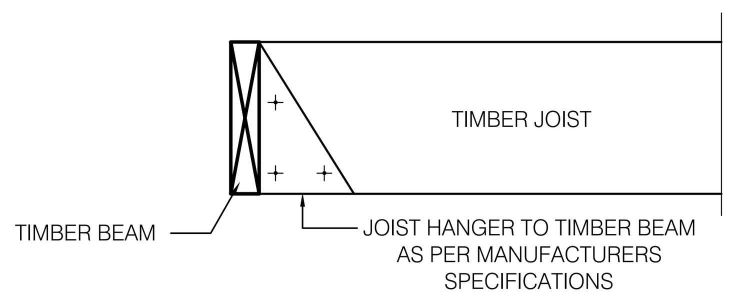
TIMBER BEAM TO BRICKWORK CONNECTION



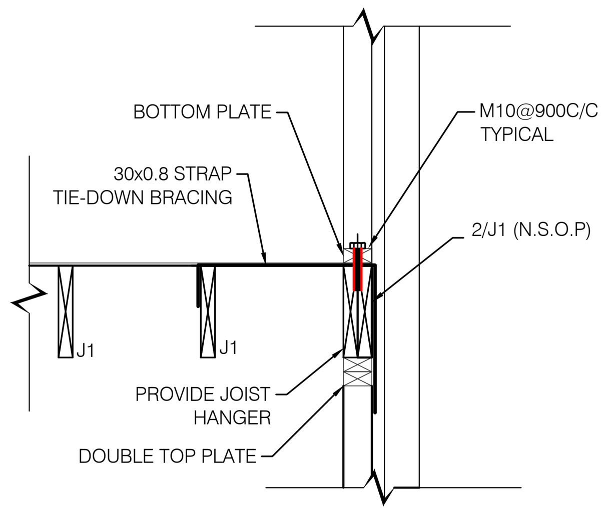
TIMBER BEAM TO POST CONNECTION



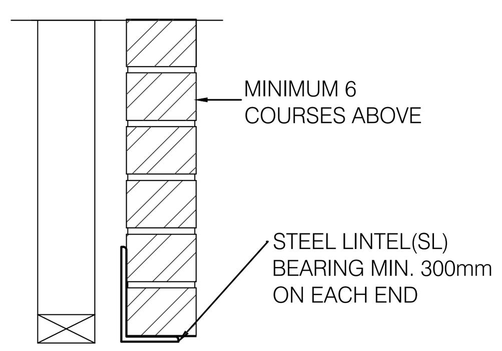
TIMBER BEAM TO POST CONNECTION



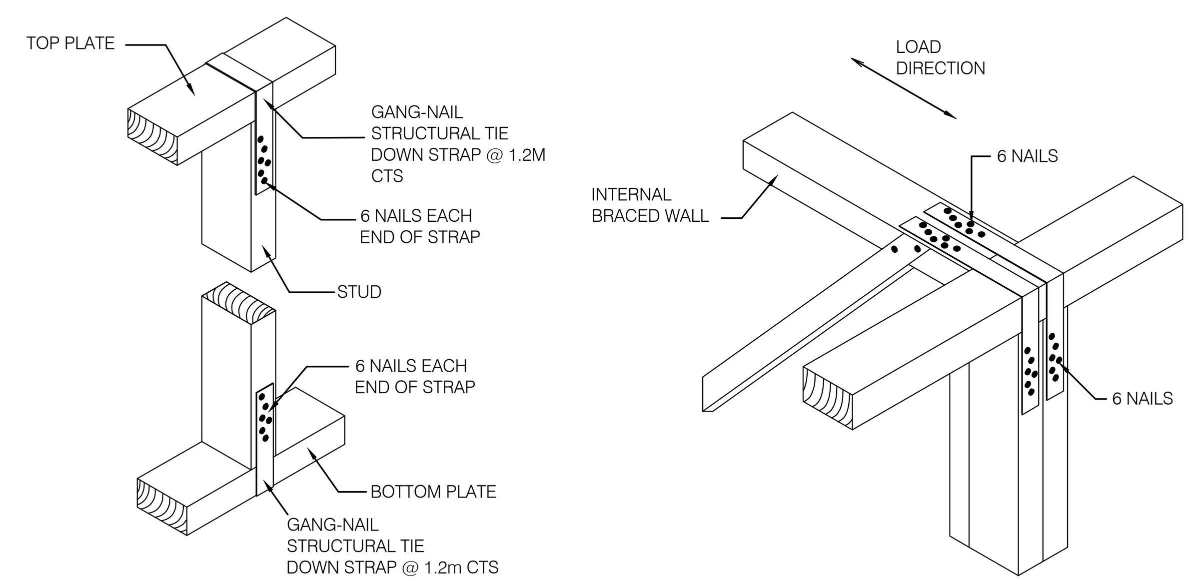
JOIST TO TIMBER BEAM CONNECTION



JOIST TO STUD WALL CONNECTION DETAIL



TYPICAL STEEL LINTEL DETAIL
SCALE 1:10



STRUCTURAL TIE DOWN STRAP

NORTH POINT

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GEMSTRUX CONSULTING ENGINEERS
LEVEL 5 SUITE 7/1C HOMEBUSH BAY DR, RHODES NSW 2138
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GEMSTRUX
CONSULTING ENGINEERS

CLIENT:
DAVID BOURCHAN

SITE:
16 HENSON STREET,
MERRYLANDS, NSW 2160

TITLE:
FRAMING DETAILS AND SECTIONS

APPROVED: J.M.
B.E CONSTRUCTION MIEAust MEMB. NO: 4468815
DESIGNED: J.M.
DRAWN: J.P.F.

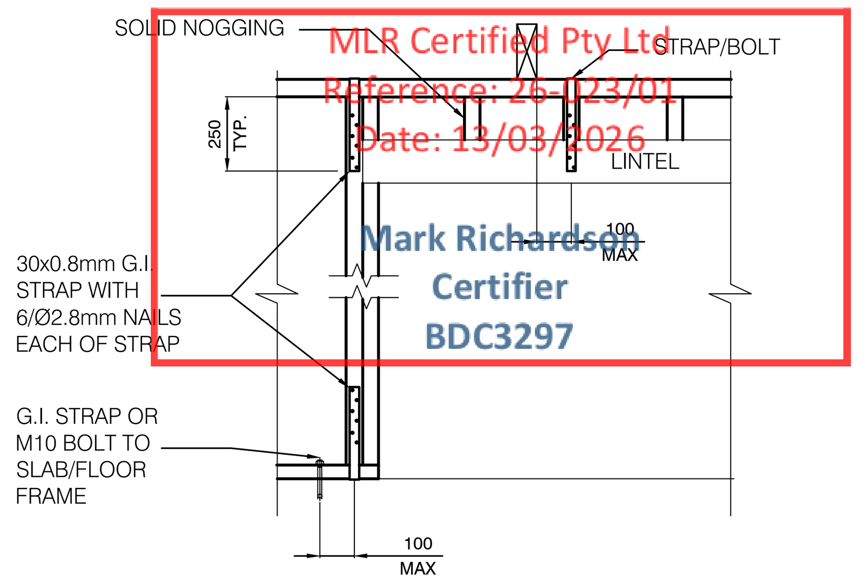
0 0.1 0.2 0.3 0.4 0.5 1.0 1.5 2.0
GRAPHIC SCALE 1:20

PROJECT NO:
G26043

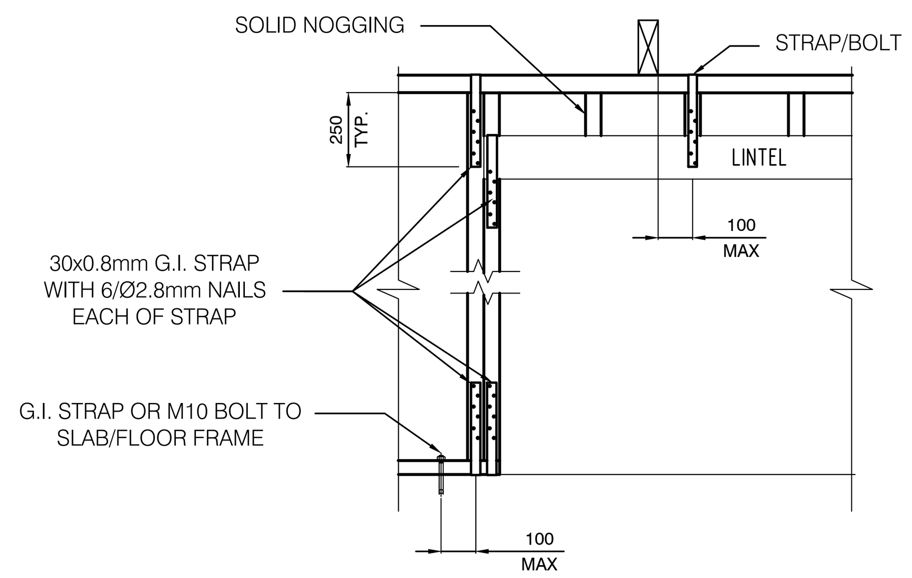
DRAWING NO:
S13

SCALE @ A1
1:20 U.N.O

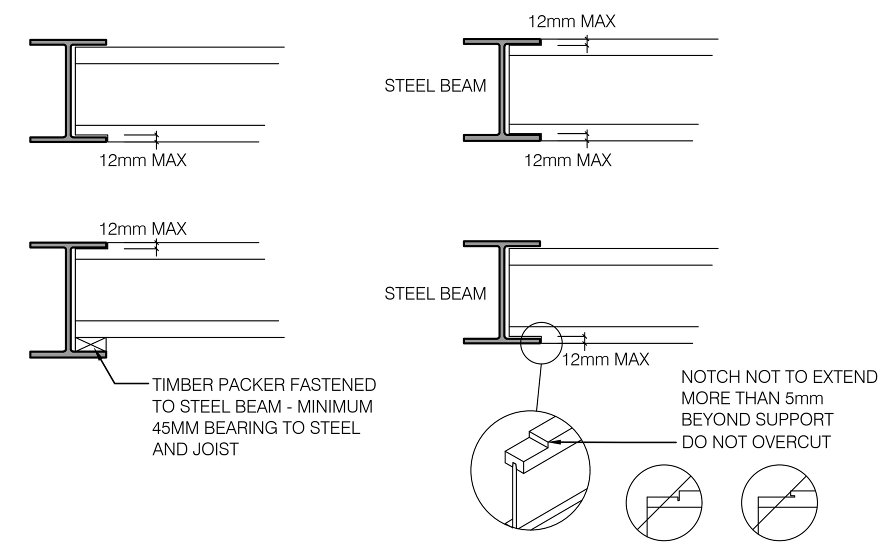
MLR Certified Pty Ltd
Reference: 26-023/01
Date: 13/03/2026
Mark Richardson
Certifier
BDC3297



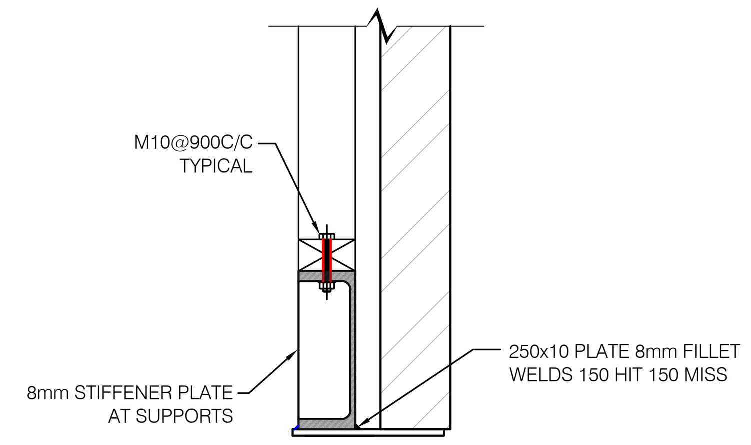
LINTEL TIE DOWN FOR OPENINGS UP TO 2000mm WIDE
SCALE: NTS



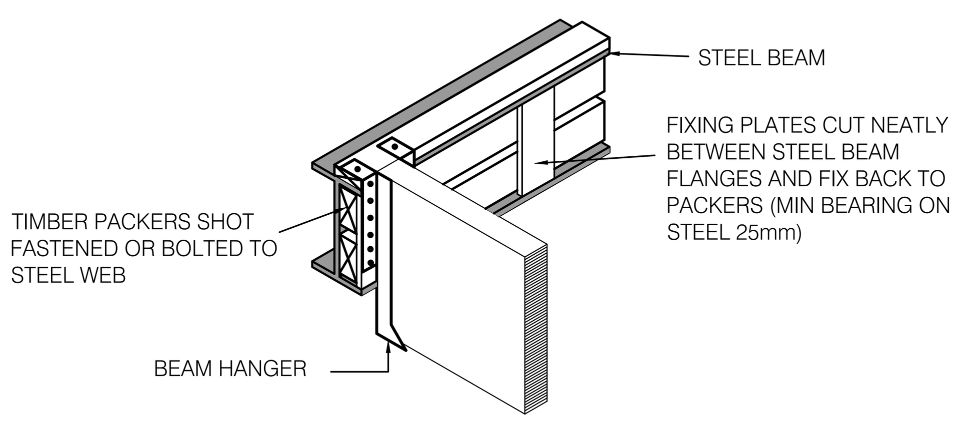
LINTEL TIE DOWN FOR OPENINGS UP TO 4000mm WIDE
SCALE: NTS



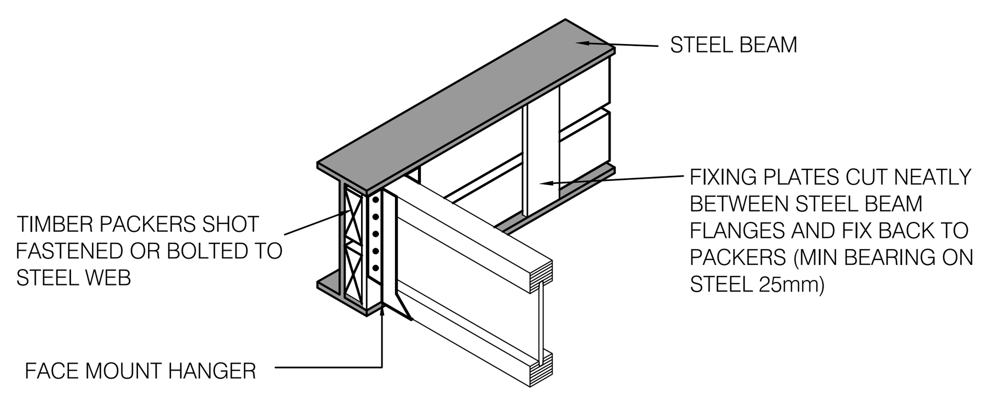
NOTCHING OF FLANGES TO END SUPPORT DETAILS
NTS



TYPICAL PFC+PLATE DETAIL



TIMBER BEAM TO STEEL BEAM CONNECTION DETAIL
SCALE 1:10



TIMBER JOIST TO STEEL BEAM CONNECTION
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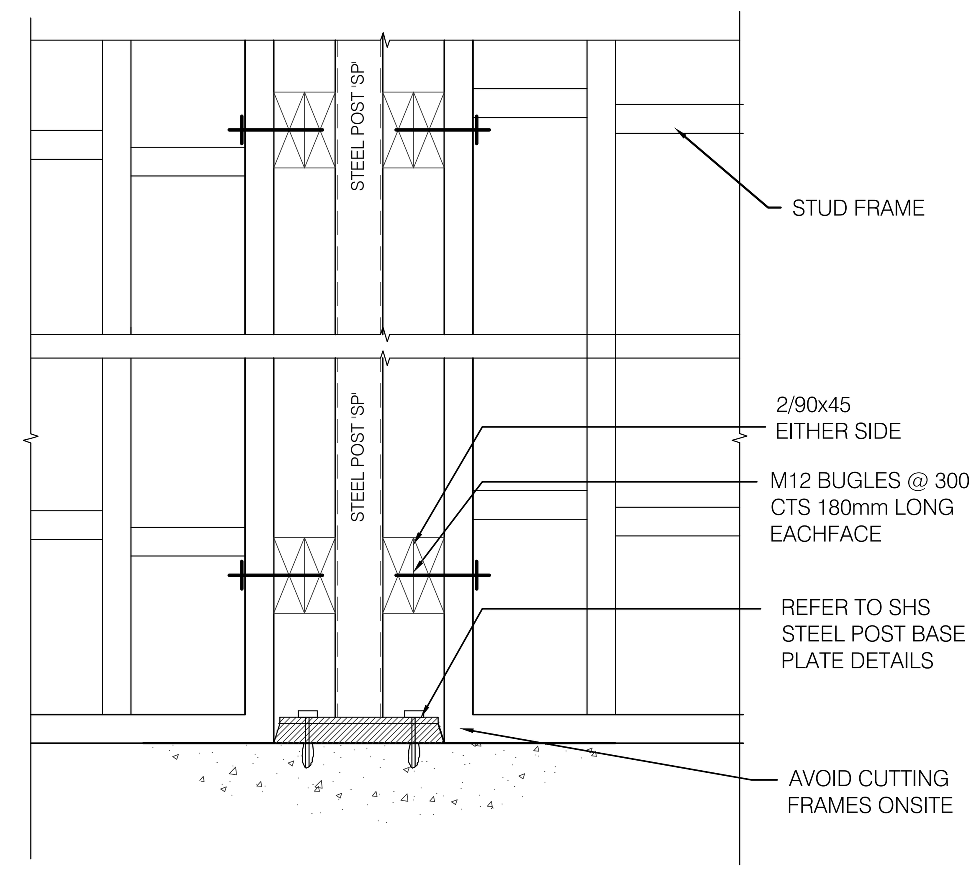
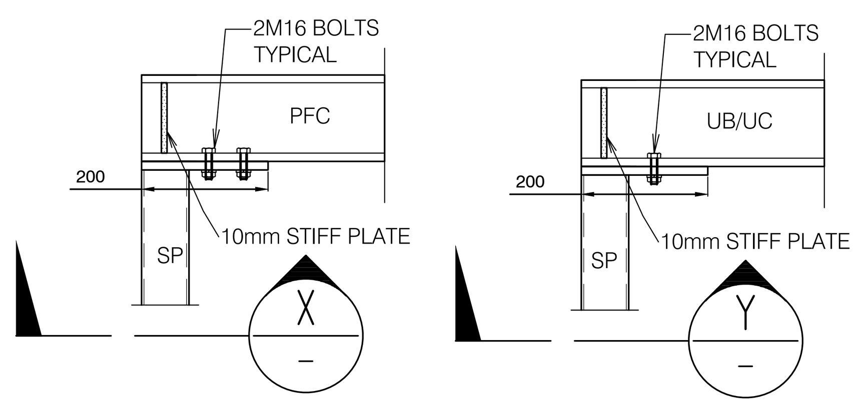
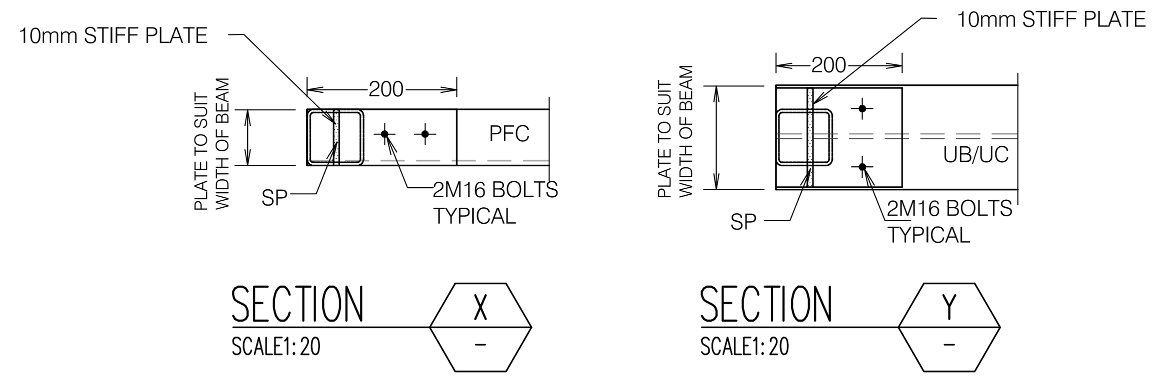


STEEL POST TO SLAB CONNECTION DETAIL
SCALE NTS



STEEL POST TO BEAM CONNECTION DETAIL



STEEL POST TO BEAM CONNECTION DETAIL

NORTH POINT

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REVISIONS

Rev	Date	Description	Verified
REV A	13/02/2026	ISSUED FOR COORDINATION	J.M.

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DRAWN: J.P.F.

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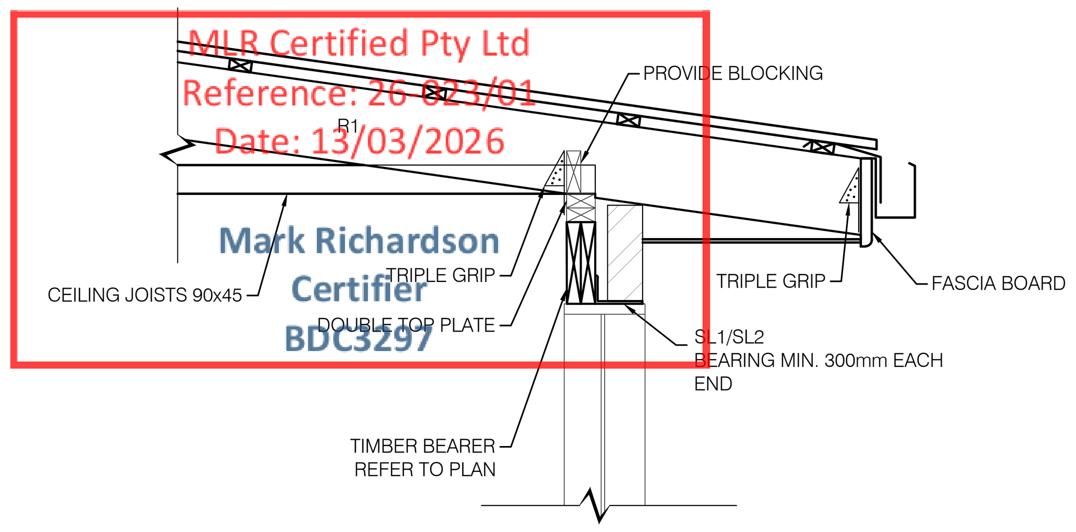
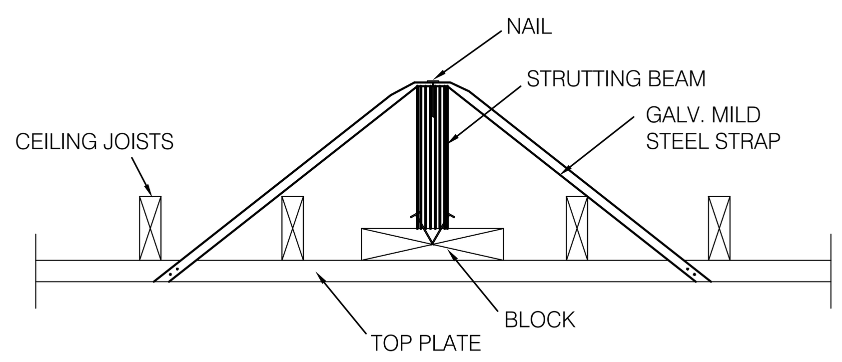
GRAPHIC SCALE 1:20

PROJECT NO: G26043
DRAWING NO: S14
SCALE @ A1: 1:20 U.N.O



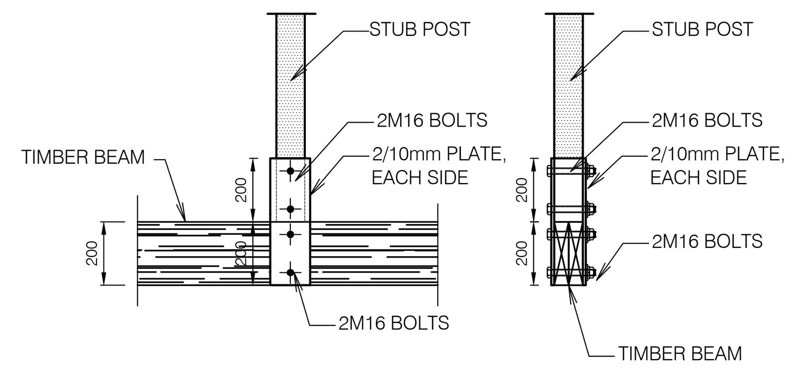
TIMBER BEARER WINDOW LINTEL - TYPICAL DETAIL

SCALE 1:20

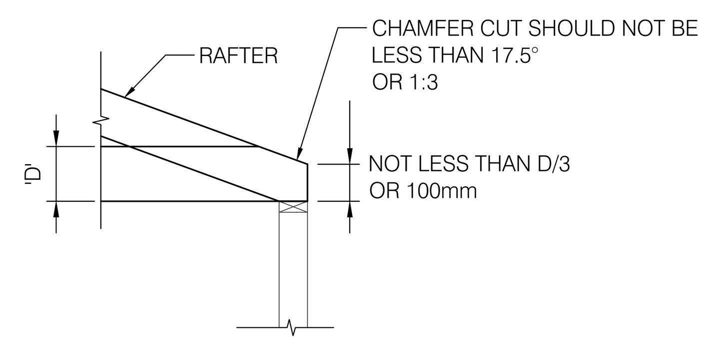


END RESTRAINT DETAIL FOR STRUTTING BEAMS

NTS

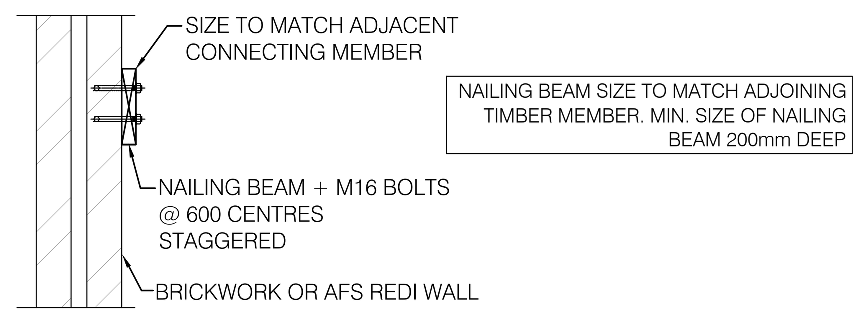


TIMBER POST TO TIMBER BEAM CONNECTION

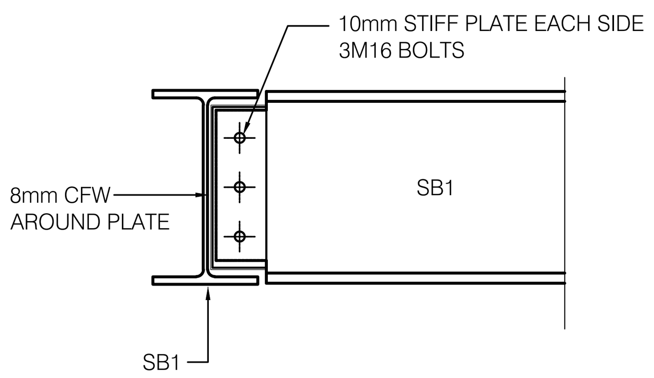


CHAMFER OF TIMBER LVL AT SUPPORTS

SCALE 1:20

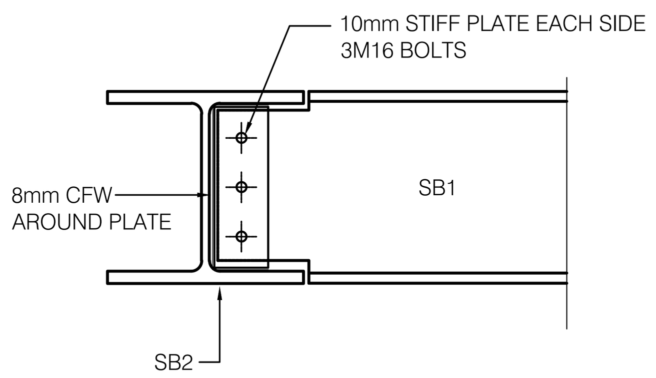


NAILING BEAM DETAIL



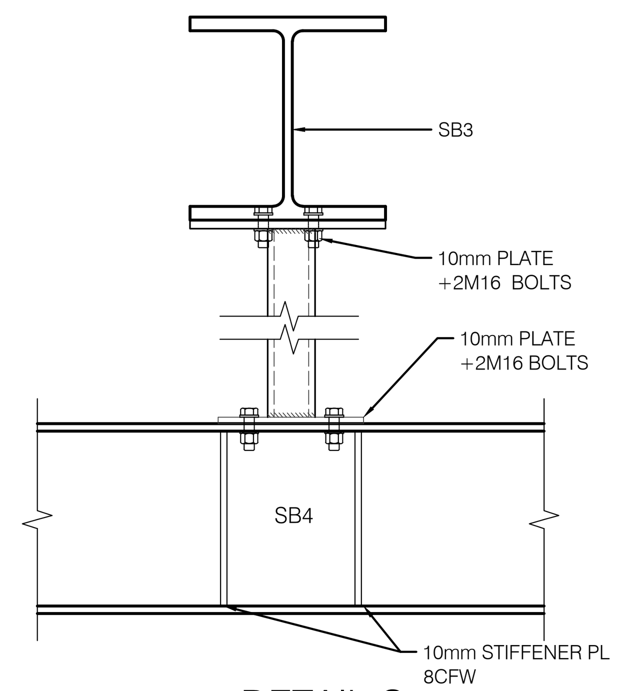
DETAIL A

SCALE 1:10



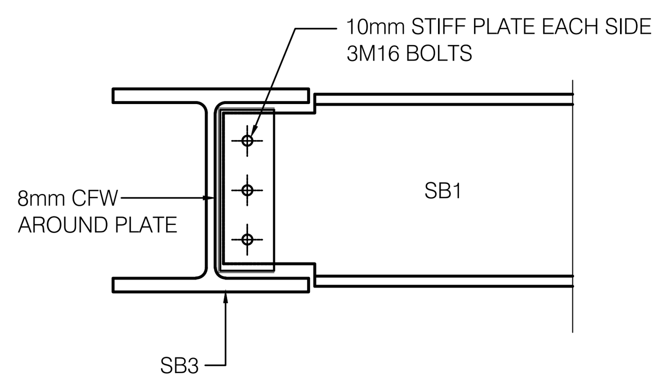
DETAIL B

SCALE 1:10



DETAIL C

SCALE 1:10



DETAIL D

SCALE 1:10

NORTH POINT

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PROJECT NO:
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S15

SCALE @ A1
1:20 U.N.O